WEEKLY ROLLUP • PhD RESEARCH WATCH

Particulate Matter
Week in Review

Week ending 24 October 2026 | 18 October 2026 to 24 October 2026 | 119 records pooled from 7 daily issues

119
RECORDS POOLED
(119 UNIQUE DOIs)

39%
MEASUREMENT SIDE
(46 RECORDS)

10
SUBTOPICS
ACTIVE

30
POOLED EFFECT
ESTIMATES (30 WITH A
CI)

25
ABSTRACT-FREE
(21% OF WEEK)

The week in one paragraph

Volume steady at 119, tier A at a series high of 18 %, and the week's epidemiology split $PM_{2.5}$ into its parts: sulfate, nitrate and ammonium carried the cardiorenal, cognitive and atrial-fibrillation signals, and elemental carbon carried ADHD. On the measurement side, a meta-analysis put a number on the optical-monitor bias every calibration inherits.

Scale. 119 records over seven issues on daily totals of 13, 18, 19, 18, 17, **24** and **10**. Against recent rollups: 119 after 123, 87, 125, 162, 148 and 126. **All ten subtopics are populated** for the fifth week running.
The measurement side held at 39 %, but its make-up flipped. *Exposure assessment & modelling* 26 → **31** (forecasting emulators, CTM and emission-factor work); *Sensing* 22 → **15**. Share series 44, 33, 42, 44, 39, **39 %**.
Cardiovascular & metabolic led the health side's growth: 7 → **12**; *Respiratory* 8 → **11**; *Burden* 18 → 14 and *Mechanistic toxicology* 15 → 10 contracted.
Estimates: 30, every one with an interval (fifth complete week at 100 %). One per 4.0 records. 27 sit on ratio scales (HR 16, RR 6, OR 5) and 3 on difference scales (FEV_1 in mL, HAZ, and a log-ratio of optical to gravimetric mass).
Abstract gap 25 of 119 (21 %), unchanged from last week. **Tier mix 21 A, 43 B, 55 C** — tier A at **18 %**, up from 12 %.

Subtopic trend and inflection

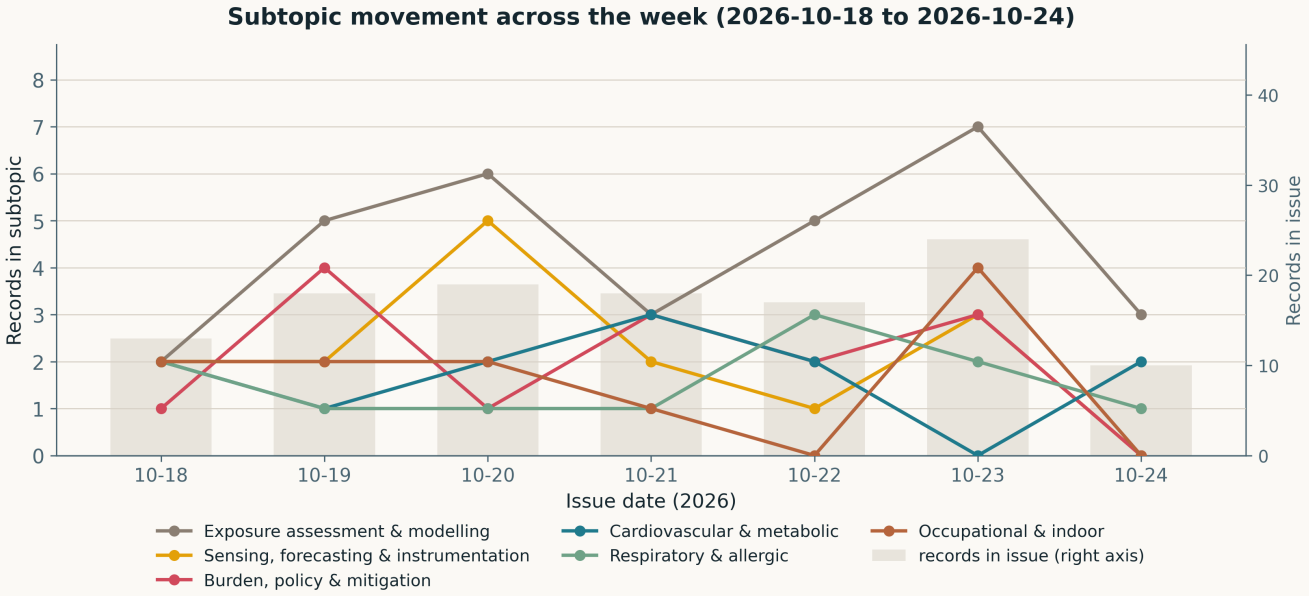


Fig. Per-issue subtopic counts across the week from `state/metrics.csv`, seven issues. Daily totals 13, 18, 19, 18, 17, 24 and 10. 18 October was built on the 19 October run, 19–22 October on their next-day runs; the scheduled runs for 23–29 October did not complete, so 23 and 24 October and this rollup were built on the **30 October recovery run**.

Expanding. *Exposure* 26 → **31** is the largest absolute gain and is forecasting-heavy: a physics–AI coupled CTM forecaster over 338 cities, a km-scale generative global $PM_{2.5}$ forecaster (Foehn), and a GEOS-CF emulator whose skill collapsed under blocked spatial cross-validation (RF on latitude/longitude alone R^2 0.91; r 0.11 on an unseen sector). *Cardiovascular & metabolic* 7 → **12** carries the week's largest cohorts.

Contracting. *Sensing* 22 → **15**, *Mechanistic toxicology* 15 → 10, *Burden* 18 → 14. *Neuro* flat at 4, but two of the four are tier A or carry components (Danish ADHD; CLHLS cognition).

The inflection worth naming. Composition displaced mass as the exposure that carries the effect. In one week: incident eGFR decline in 307,639 adults at HR **1.273** (sulfate), **1.304** (nitrate) and **1.280** (ammonium); cognitive disorders in CLHLS older adults at HR 1.030–1.033 for the three secondary inorganic ions against **1.005** for $PM_{2.5}$ mass; incident atrial fibrillation in 26.3 million Medicare beneficiaries at HR **1.056** per unit sulfate-source $PM_{2.5}$ vs **1.029** for vehicle $PM_{2.5}$; and a second-year-of-life ADHD window that follows **elemental carbon** (HR 1.24) and is unchanged after $PM_{2.5}$ adjustment. Every component exposure here is modelled (CTM, satellite–ML or source apportionment), not measured. Last week this channel was the smoke carbonaceous fraction; this week it is secondary inorganics. **The test for next week**: a component-resolved estimate built on measured speciation, or a network-scale absorption (BC) channel used as the exposure.

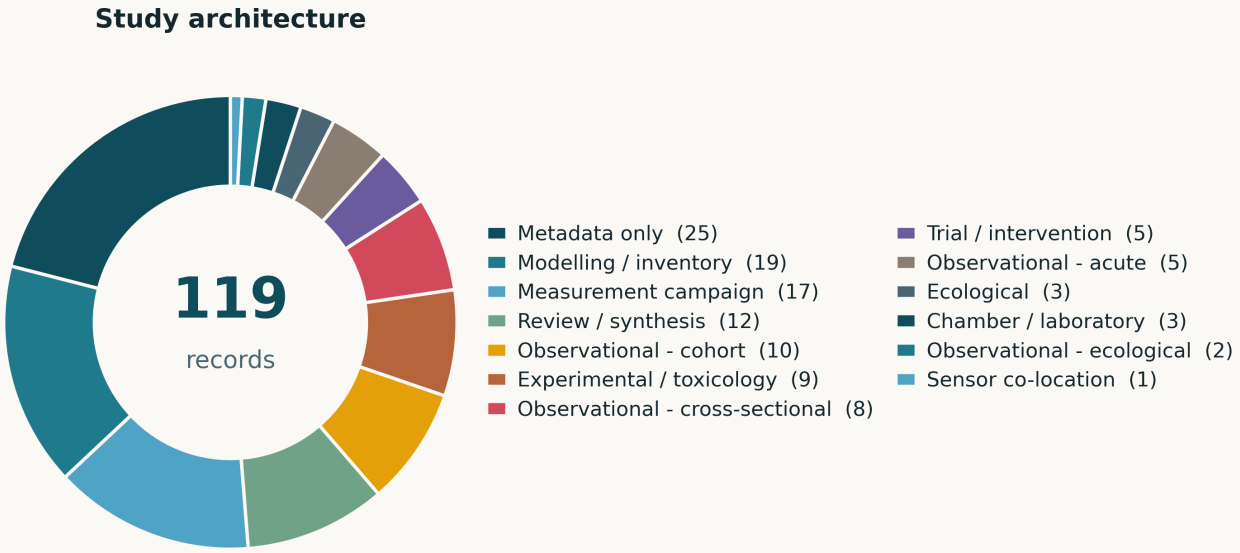


Fig. Study architecture over the pooled week. *Metadata only* 25 leads, then *Modelling / inventory* 19, *Review / synthesis* 12, *Measurement campaign* 12 and *Observational – cohort* 10. **Three interventional records**: the HEPA vs sham crossover (cytokines, visceral adiposity, 18 October), a traffic-noise/air-pollution randomised crossover (lung function), and the HAPIN LPG follow-up to age 5 (null for HAZ).

Pooled analytics

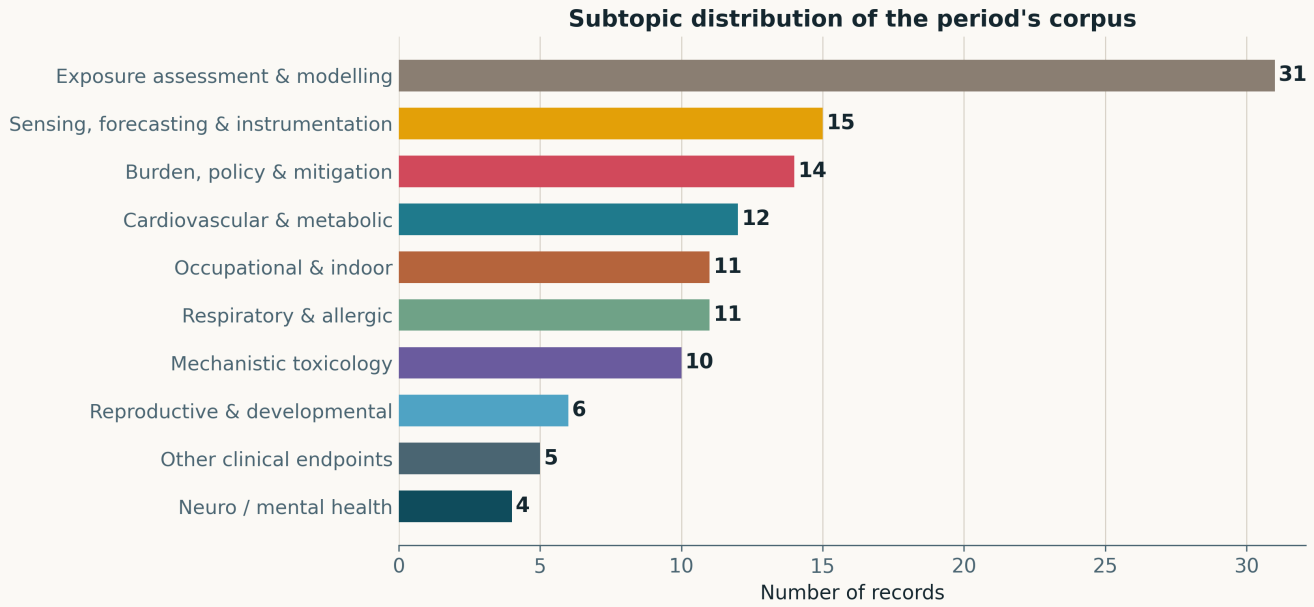


Fig. Pooled subtopic distribution, all ten bands populated: *Exposure* 31 and *Sensing* 15 put the measurement side at **46 of 119 (39 %)**. On the health side *Burden* 14 and *Cardiovascular* 12 lead, then *Respiratory* and *Occupational & indoor* 11 each, *Mechanistic toxicology* 10; *Neuro* is smallest at 4.

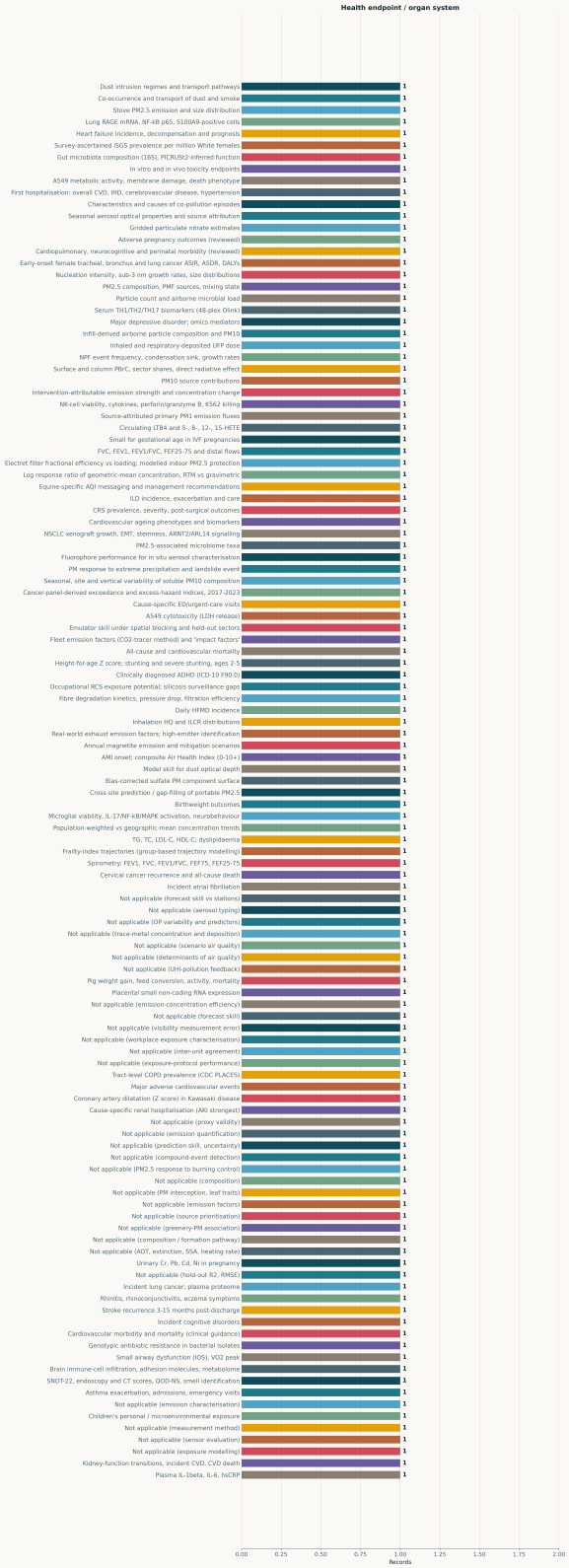


Fig. Health endpoint over the pooled week, canonicalised. "No health endpoint" pools the metadata-only records with instrument, forecasting, emissions and modelling work. Among human outcomes, renal, cognitive, cardiac-rhythm and mortality endpoints lead; antimicrobial resistance and cervical-cancer recurrence are new endpoint classes for the watch.

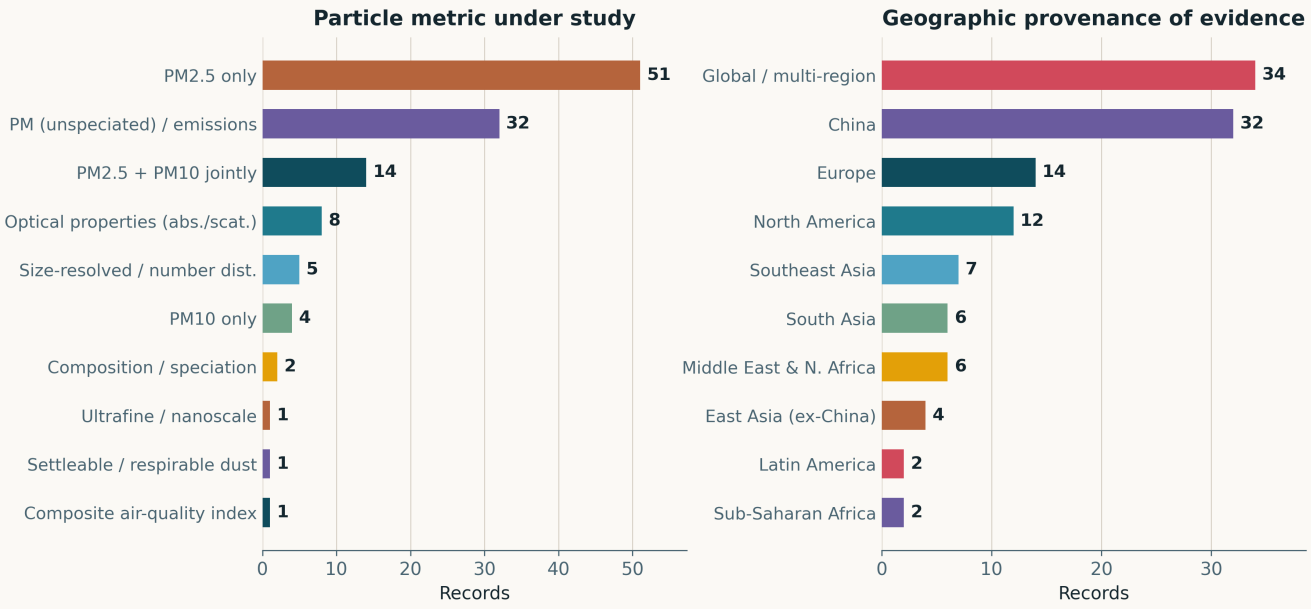


Fig. Left — particle metric: **$PM_{2.5}$ alone 51 of 119**, unspeciated PM or emissions 32, joint $PM_{2.5}$ + PM_{10} 14, optical properties 6, size-resolved 5, PM_{10} alone 4, composition 2. The component-resolved epidemiology of the week is coded under $PM_{2.5}$, so the composition bar understates it. Right — geography: *Global / multi-region* 34 (reviews, laboratory work, the six- and 102-country records), **China 32**, Europe 14, North America 12, Southeast Asia 7, South Asia 5, MENA 4, East Asia (ex-China) 4.

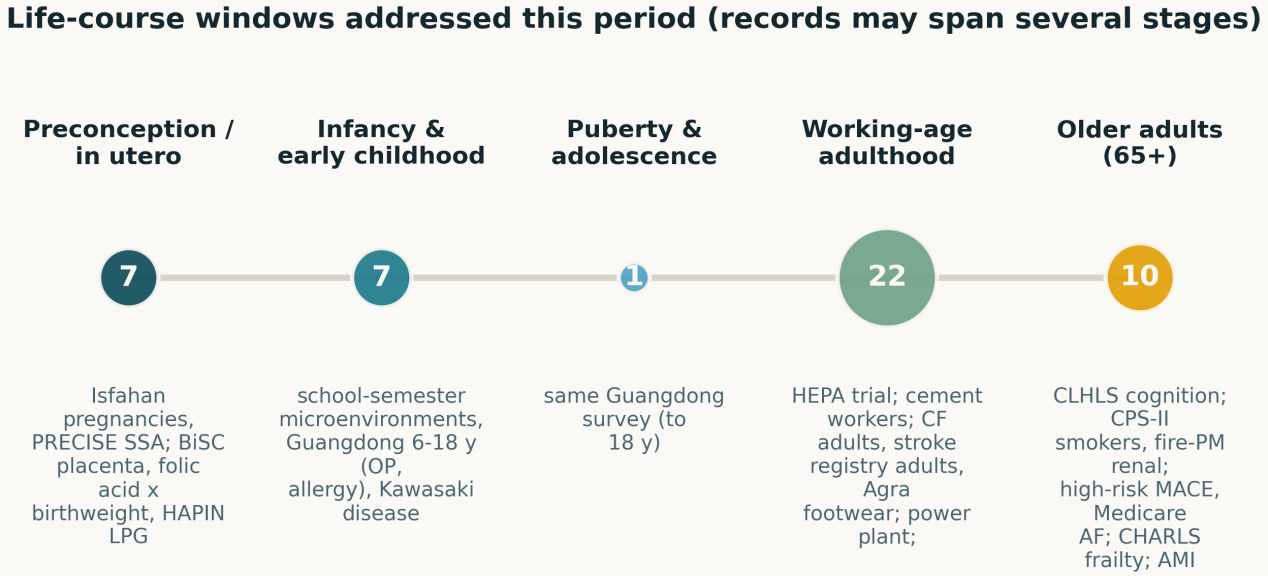


Fig. Life-course windows across the pooled week; counts non-exclusive. **Working age 22**, older adults 10, preconception / in utero 7, infancy & early childhood 7, **adolescence 1** — the adolescent window fell back to a single record after last week's 4.



Fig. The **27** ratio-scale estimates, every one with a 95 % CI. **Not mutually comparable and not meta-analysed**: increments include per 1 $\mu g\,m^{-3}$, per IQR, quartile and binary contrasts, a unitless trajectory index and an MR instrument, on HR, OR and RR scales, so no pooled estimate or I^2 is reported. Within-family heterogeneity is low where it can be read: the three cardiorenal ion HRs (1.273–1.304) and the three CLHLS ion HRs (1.020–1.033) overlap almost entirely, consistent with collinear modelled components rather than independent effects. **Two intervals cross unity**: *S. enterica* tetracycline resistance RR 1.65 (0.98–2.76) and SO_2 stroke recurrence HR 1.19 (0.94–1.50). The Beijing per-1 $\mu g\,m^{-3}$ mortality HR (1.053) is an order of magnitude above pooled long-term estimates. Nested families share cases: cardiorenal 3, CLHLS 4, Medicare AF 2, cervical cancer 2, Danish ADHD 2, Beijing 2, Guangzhou SGA 3, Medicare smoke waves 2, antimicrobial resistance 2.

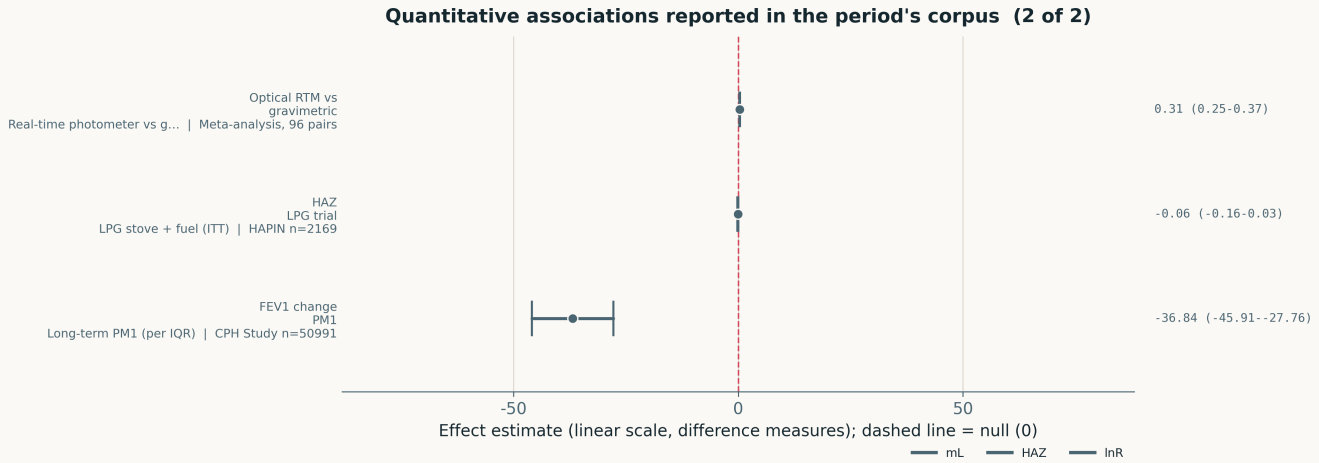


Fig. The **3** difference-scale estimates: FEV_1 −36.8 mL (−45.9 to −27.8) for PM_1 in 50,991 adults; HAPIN ITT HAZ −0.06 (−0.16 to 0.03, null); and the meta-analytic log-ratio of real-time optical monitors to gravimetric mass, **0.31** (0.25–0.37), i.e. optical instruments read about **36 %** high across 96 paired comparisons.

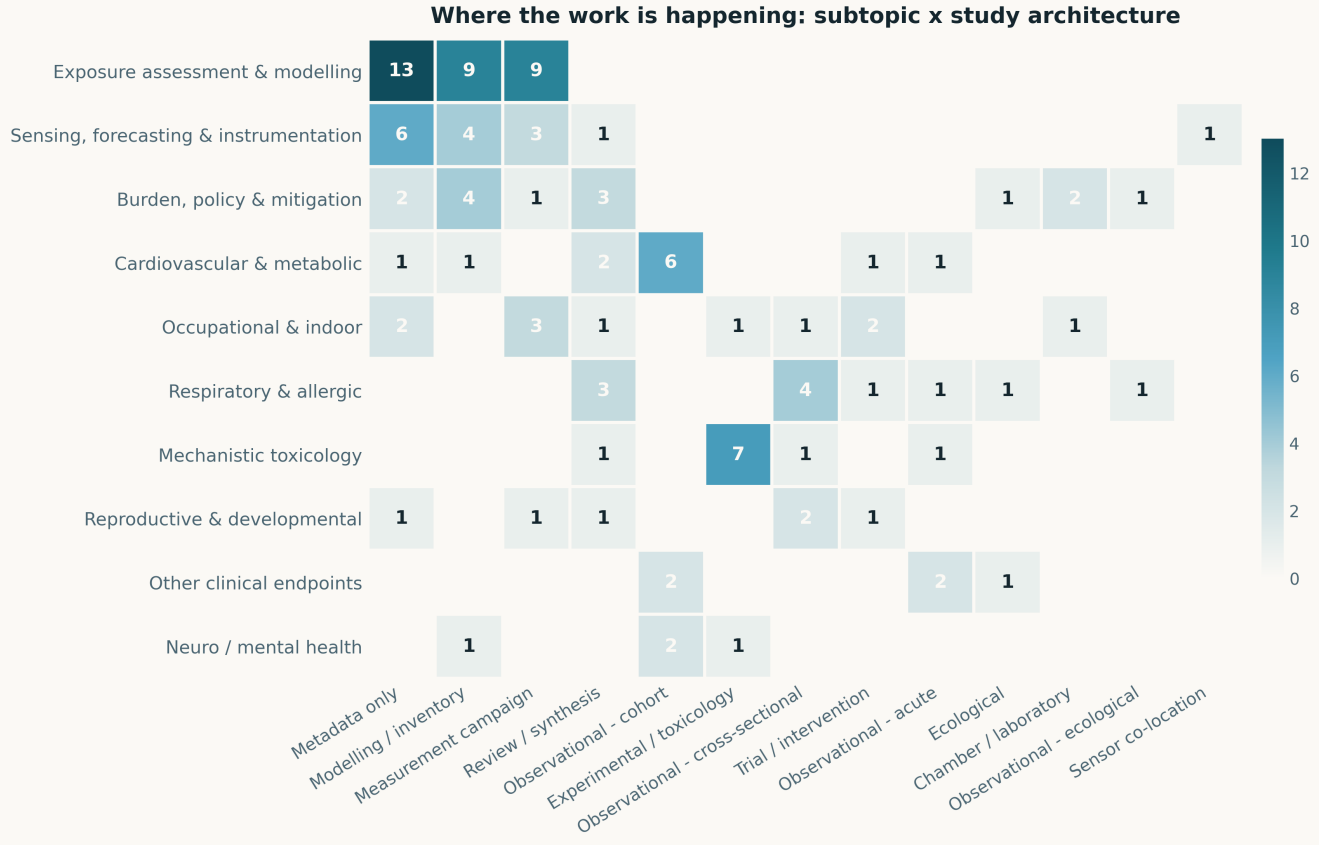


Fig. Subtopic × architecture, pooled. The densest cells are *Exposure* × *Metadata only* and *Exposure* × modelling; cohorts concentrate in *Cardiovascular* and *Neuro*; sensing is split between instrument evaluation and field measurement.

Venues, trial watch, and what the pipeline did

Venues. *Environ Sci Technol* leads at **11** (9 %), then *Atmos Environ* 8, *Atmosphere* 6, *Research Square* 6, *ACS Environ Health* 5, and *Atmos Chem Phys* and *Atmos Pollut Res* at 4. **Preprints supplied 9** (Research Square 6, among them medRxiv with the HAPIN follow-up, arXiv with Foehn).

Recurring authors and most-cited work. Surname frequency remains a naming artefact; **no repeat-group cluster is claimed**. Citation counts for records days old are uninformative; the largest datasets — 65.2 and 26.3 million Medicare beneficiaries, 12.5 million admissions across six countries, 397,103 bacterial isolates from 102 countries — each supplied one record.

Trial watch. ClinicalTrials.gov `LastUpdatePostDate` windows 19–20, 20–21, 21–22 and 22–23 October: 0, 0, 2 and 2 hits. Stored: AIRWEIGHS NCT02763917 (HEPA purifier RCT, update) and the Taif COPD prevalence survey NCT07886346 (observational, low relevance); APCIQ-BP NCT05903950 updated. `trials.json` held **19 trials** over **35 windows** at period close. (A new controlled woodsmoke-exposure study, NCT07536178, surfaced on the 30 October sweep and belongs to W44.) The week's interventional evidence again came from the literature — three crossover/RCT records — not the registry.

What the harvesting legs did. Europe PMC lagged hard mid-week (3, 53, 1 records on 20–22 October) and then re-surfaced PubMed records one to two days late, producing many cross-day duplicates on 23–25 October, all caught at screening or by the DOI gate. **Crossref-by-ISSN** 7–69/day. **PubMed connector** appended 2–20 records/day. **OpenAlex** HTTP 429 throughout. **arXiv** supplied Foehn, held one day because its DataCite DOI was not yet registered. **Consensus** 10 hits, **0 new**; the DOI gate caught one Consensus duplicate (Ginsburg 2026, shipped 29 July). **Scholar Gateway** not called.

Open gaps

- **The optical bias now has a pooled number and still no home.** Real-time optical monitors read ≈36 % above gravimetric (log-ratio 0.31, 0.25–0.37); no health record in the week propagates that bias, or calibration uncertainty, into an exposure–response estimate.
- **Component effects rest on modelled components.** Every sulfate/nitrate/ammonium and source-specific estimate this week comes from CTM, satellite–ML or receptor-model output; none from measured speciation at the subject's location.
- **Forecast skill is over-reported under random CV.** The GEOS-CF emulator's collapse under a 300 km spatial block is a general warning for gridded PM products used to calibrate or gap-fill networks.
- **The abstract gap is flat at 21 %**, concentrated in day-0 Elsevier deposits.
- **Held queue empty.** The 3 October ES&T NK-cell record entered the 23 October issue; Foehn was released on 21 October.
- **Pipeline defects carried forward**: scheduled runs for 23–29 October did not complete (recovered on 30 October); Europe PMC late re-surfacing makes per-day dedup order-dependent; OpenAlex dead; Scholar Gateway not reconnected; `.git` lock debris.

Paper-level digest — the week by subtopic

Every record that appeared in a daily issue keeps that issue's critical summary verbatim; nothing here is re-written or re-assessed. Records surfaced without an abstract are shown as such. Ordered by subtopic, then by the date of the issue that first carried the record.

Sensing, forecasting & instrumentation — 15 records

Carried from the daily issues — full summaries (15 of 15 records in this subtopic)

Birinci Bosphorus bridge traffic ML proxy $PM_{2.5}$ NO_2 2026 Atmosphere

Machine-Learning Proxy of Air-Quality Variability Associated with Intercontinental Bridge Traffic over the Bosphorus in Istanbul

17,520-h panel (2022-2023) pairing six bridge traffic sensors with the nearest monitors. A Prophet traffic-only baseline gave negative R2 for every bridge x pollutant. LightGBM, XGBoost and a BiLSTM-attention model with meteorology, calendar, precipitation lags and short autoregressive terms reached mean hold-out R2 ~0.77 (0.58-0.86), RMSE down 47-63%; the three families were practically equivalent. SHAP ranked autocorrelation, season, wind direction and heavy vehicles above total traffic. Limitation: autoregressive pollutant terms make this nowcasting persistence, not a traffic proxy - the headline R2 largely reflects lagged concentrations, and 'closest monitor' distance is unstated. Relevance: moderate - shows traffic counts alone do not substitute for measurement in complex coastal flow.

doi: 10.3390/atmos17101024

Cai self-supervised stacking particulate nitrate sparse observations 2026 Atmos Environ

A Self-supervised Stacking Framework for Estimating Particulate Nitrate in Two Chinese Urban Agglomerations under Sparse Ground Observations

METADATA ONLY. Title and authors only (Cai, He, Li, Liu et al.). No abstract was retrievable from Crossref, Europe PMC or Semantic Scholar at harvest. The title describes a self-supervised stacking ensemble that estimates particulate nitrate in two Chinese urban agglomerations from sparse ground observations. Predictors, validation design and skill cannot be stated. Limitation: unverifiable beyond the title; with sparse speciation sites, random cross-validation would overstate skill (see the 22 Oct blocked-CV record). Relevance: medium - component estimation under sparse reference sites is the same problem as filling between LCS anchors.

doi: 10.1016/j.atmosenv.2026.122433

Cover solvatochromic fluorophores in situ aerosol fluorescence 2026 Aerosol Sci Technol

Evaluation of solvatochromic fluorophores for in situ fluorescence characterization of aerosols

METADATA ONLY. Title and authors only (Cover, Gibbons, Boadu, Ohno et al.). No abstract was retrievable from Crossref or Semantic Scholar at harvest. The title describes evaluating solvatochromic dyes, whose emission shifts with local polarity, as in-situ probes of aerosol particle properties. Analytes, particle types and performance metrics cannot be stated. Limitation: unverifiable beyond the title. Solvatochromic readouts also confound polarity with viscosity and water content, so any phase-state inference needs independent calibration. Relevance: low-medium - a laboratory composition probe, not a field-deployable sensor channel.

doi: 10.1080/02786826.2026.2746515

Ghaffarpasand Pope vehicle remote sensing real-world emissions high emitters 2026 Atmos Pollut Res

Emerging Evidence on Real-World Exhaust Emissions and High Emitting Vehicles from Remote Sensing Observations

METADATA ONLY. Title and authors only (Ghaffarpasand, Pope); no abstract at harvest. Appears to synthesise vehicle emission remote-sensing (VERSS) evidence on real-world exhaust and high emitters. Whether PM/smoke opacity or only NOx/CO/HC is covered cannot be stated. Limitation: unverifiable beyond the title; VERSS PM is inferred from UV opacity with large per-pass uncertainty, and single-pass snapshots overweight hot-running conditions. Relevance: moderate - high-emitter fractions explain the kerbside spikes roadside LCS record but cannot attribute.

doi: 10.1016/j.apr.2026.103244

Huang foliar particulate UV-Vis-NIR calibration transfer 2026 Environ Monit Assess

Nondestructive UV-Vis-NIR assessment of foliar particulate retention in urban vegetation: cross-species transfer and calibration correction

Reflectance models built on 180 Ligustrum leaves and transferred to 45 Photinia leaves from park, street and industrial zones. Zone classification reached 91.1% after excluding endpoint and water bands. Direct mass prediction on the target species failed (R2 0.265, positive bias); a linear output correction using 15 target reference leaves (five per zone), tested on the other 30 over 50 splits, gave R2 0.797 +/- 0.042. Limitation: target-based scheme selection and the correction set share zones with the test set, and deposit properties are unmeasured, so this is reference-assisted local calibration, not transfer. Relevance: moderate - the same slope-and-offset recalibration logic, and the same failure of naive transfer, governs moving LCS calibrations between sites.

doi: 10.1007/s10661-026-16062-x

Intra four co-deployed electrostatic PM monitors power plant agreement 2026 Research Square (preprint)

Multi-point monitoring of PM_{10}, $PM_{2.5}$, and PM_1.0 at a combined-cycle power plant: field agreement between four co-deployed electrostatic monitors

Four prototype three-channel electrostatic mass monitors at a combined-cycle plant, Feb-Jun 2020, 92-100% daily recovery; station means PM_{10} 44.9-54.1, $PM_{2.5}$ 24.8-30.8, PM_1.0 7.5-11.9 $\mu g\,m^{-3}$. Fixed station differences explained 0.7% of log 24-h PM_{10} variance but the day-by-station interaction 57.1%; pairwise 95% limits of agreement spanned ratios 0.35-3.26 (PM_{10}) and 0.33-2.83 ($PM_{2.5}$); single-unit ICC 0.42 for PM_{10}, 0.75 for the four-unit mean. Disagreement did not shrink with averaging. Limitation: preprint; no reference instrument, so agreement says nothing about accuracy; units were not co-located side by side first, so unit drift and real micro-gradients are separated only by inference. Relevance: high - the same argument applies to any LCS network: station contrasts need routine co-location checks.

doi: 10.21203/rs.3.rs-10901106/v1

Kheder Foehn generative km-scale global $PM_{2.5}$ forecast 2026 arXiv (preprint)

Global probabilistic forecasting of air pollution

Foehn, a generative model trained on global kilometre-scale $PM_{2.5}$ estimates that also computes air-mass origin. In one year of retrospective forecasts it beat the operational forecast at three of four stations; its ensembles are reported as calibrated, and the ensemble mean detects local pollution extremes four days ahead about as well as the operational system does one day ahead. Limitation: preprint; 'kilometre-scale estimates' are themselves satellite/model fusion products, so skill is partly measured against the training distribution, and the authors concede real-time km-scale inputs do not yet exist - operational use is hypothetical. Skill metrics and station counts are not given in the abstract. Relevance: high - calibrated LCS networks are the plausible real-time km-scale input this model says it lacks.

doi: 10.48550/arXiv.2610.25941

Li runway visual range model brown carbon aerosol chromatic effects 2026 Opt Lett

Refined runway visual range measurement model incorporating brown carbon aerosol effects

Optics Letter: conventional runway visual range (RVR) algorithms treat black carbon and scattering aerosol only; brown carbon's wavelength-selective absorption introduces two systematic errors - extinction spectral mismatch between the transmissometer/forward-scatter wavelength and the photopic band, and a chromatic shift in the visual threshold. A refined RVR model including BrC chromatic effects is proposed and 'experimentally validated'. Limitation: no error magnitudes, BrC loadings or validation data in the abstract. Relevance: moderate - the same single-wavelength assumption underlies optical PM sensors, whose mass conversion is biased by absorbing, spectrally selective aerosol.

doi: 10.1364/ol.611004

Pyae Tin Saw portable $PM_{2.5}$ cross-site prediction fixed station 2026 Environ Pollut

Short-term $PM_{2.5}$ data continuity: Cross-site prediction of portable measurements using a nearby fixed monitoring station

METADATA ONLY. Title and authors only (Pyae Tin Saw, Kallawicha); day-0 Elsevier deposit with no abstract anywhere at harvest. The title describes predicting short-term portable $PM_{2.5}$ records from a nearby fixed monitoring station to maintain data continuity. Sensor model, distance, n, and skill metrics cannot be stated. Limitation: unverifiable beyond the title; cross-site transfer functions are typically valid only within the regional-background regime and fail for local sources, which is exactly when portable data matter. Relevance: high - gap-filling an LCS or portable record against a regulatory anchor is a core network-operations problem.

doi: 10.1016/j.envpol.2026.129384

Ren physics-constrained UQ urban $PM_{2.5}$ prediction driving factors 2026 Build Environ

Integrating Implicit Physics Constraints and Uncertainty Quantification for Urban $PM_{2.5}$ Prediction and Driving Factor Analysis

METADATA ONLY. Title and authors only; no abstract retrievable at harvest. The title describes an urban $PM_{2.5}$ prediction model with implicit physics constraints and uncertainty quantification, plus driving-factor attribution. Architecture, city, input features, horizon, skill metrics and whether the uncertainty is calibrated cannot be stated. Limitation: 'implicit' physics constraints in ML usually mean loss-function regularisation, which does not guarantee physical consistency out of distribution; nothing beyond the title is verifiable. Relevance: moderate - calibrated predictive uncertainty is what allows LCS-fed forecasts to propagate sensor error honestly.

doi: 10.1016/j.buildenv.2026.115449

Risti Jambi PMS5003 PMS7003 vs BAM-1020 2026 Lingkar: J Environ Eng

Evaluation of Low-Cost $PM_{2.5}$ Sensor Accuracy in Comparison with Government Reference Instrument

Consensus sweep. IoT nodes with two Plantower sensors co-located with the BMKG BAM-1020 at the Sultan Thaha Jambi climatology station, December 2025-January 2026, followed by linear-regression correction. Both tracked the reference moderately (R2 0.60 PMS5003, 0.59 PMS7003) and overestimated: PMS7003 MBE 10.46, MAE 12.53, RMSE 18.02 $\mu g\,m^{-3}$; PMS5003 MBE 11.84, MAE 13.60, RMSE 19.60 $\mu g\,m^{-3}$. The authors attribute the positive bias to high humidity. Limitation: two wet-season months at one site, no RH-stratified or kappa-Kohler correction and no held-out validation of the regression, so the correction will not transfer to the dry or haze seasons. Relevance: high - an equatorial reference co-location adds to the regional-calibration evidence; the 10 $\mu g\,m^{-3}$ bias matches humid-tropics PurpleAir behaviour.

doi: 10.22373/ljee.v7i1.10530

Sharma Singapore aerosol type classification ground remote sensing reanalysis 2026 Aerosol Air Qual Res

Aerosol Types Classification in the Tropical Climate of Singapore by Integrating Ground-Based Remote Sensing and Reanalysis Datasets

METADATA ONLY. Title and authors only; no abstract retrievable at harvest. Aerosol-type classification in tropical Singapore combining ground-based remote sensing (AERONET sun photometer presumably) with reanalysis (MERRA-2/CAMS). Classification scheme, period and the share of transboundary smoke cannot be stated. Limitation: unverifiable beyond the title; columnar typing does not map one-to-one onto surface PM. Relevance: moderate - aerosol type drives optical-LCS response (hygroscopicity, refractive index) and so the calibration regime a tropical network needs.

doi: 10.1007/s44408-026-00164-9

Sim real-time dust monitor bias gravimetric meta-analysis 2026 Ann Work Expo Health

Systematic bias in real-time dust monitoring: a meta-analysis of measurement discrepancy and its implications for occupational exposure assessment.

PRISMA meta-analysis of 112 occupational studies; 96 paired RTM-gravimetric comparisons pooled in a three-level random-effects model. Professional photometers read 36% high on average: pooled lnR 0.31 (0.25-0.37), I2 78.4%, prediction interval 0.11-0.63. Factory-calibrated units read 43% high, site-calibrated units 13% high; site-specific calibration was the strongest moderator (beta -0.24, P<0.001), followed by log GSD of the size distribution (beta 0.15) and RH (beta 0.002 per %). No eligible study compared sub-$500 sensors against gravimetry. Limitation: geometric-mean ratios from heterogeneous aerosols and campaign lengths; the prediction interval never crosses zero but its width means a single uncalibrated deployment can be off by 12-88%. Relevance: high - the size-distribution and RH moderators are the same terms LCS calibration models need, now with a pooled prior.

doi: 10.1093/annweh/wxag089

Srivastav ion particle size distribution NPF Indo-Gangetic Plain 2026 Atmos Chem Phys

Ion and particle size distribution measurements in the Indo-Gangetic Plain: seasonal insights and challenges in new particle formation event characterization

Ion and particle spectrometer measurements at a suburban IGP site, May-December 2023. NPF occurred on 30/35 valid days (86%) in May-June but 20/73 (28%) in winter/post-monsoon. Morning median condensation sink was lower on event days (0.0095 vs 0.0292 s-1). Growth rates rose with size, 14.7 nm/h (3-7 nm) to 19.0 nm/h (7-20 nm). Pollution plumes intermittently masked or distorted events, so both visual and automated classification and growth-rate fitting were unreliable. Limitation: no precursor gas (H2SO4, amines, HOMs) measurements, so mechanism is unconstrained; eight months at one site. Relevance: medium - in high-CS settings the UFP number seen by condensation or diffusion-charger sensors is plume- not nucleation-driven.

doi: 10.5194/acp-26-14917-2026

Wu ProNet physics-AI coupled CTM $PM_{2.5}$ O_3 forecasting 338 cities 2026 Natl Sci Rev

Fusing physics and AI enhances accuracy and diagnostic capabilities in air quality forecasting

ProNet embeds observation-constrained neural correctors inside the governing equations of a chemical transport model rather than post-processing its output. On 2022-23 data across 338 Chinese cities it beat the physical CTM and AI-only baselines, generalised better to extreme events outside training, and surpassed forecaster-adjusted operational forecasts by 6.1% ($PM_{2.5}$) and 3.0% (O_3). The correctors yield process-level error bookkeeping that exposes systematic CTM parameterisation biases. Limitation: '6.1% accuracy' is an unspecified metric; one two-year period invites temporal leakage, and process attribution from learned correctors is not unique - compensating errors can be assigned to the wrong process. Relevance: high - in-equation assimilation is the natural slot for dense calibrated LCS observations.

doi: 10.1093/nsr/nwag570

Exposure assessment & modelling 31 records

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (31 OF 31 RECORDS IN THIS SUBTOPIC)

Alturki ensemble ML emulation GEOS-CF $PM_{2.5}$ Arabian Peninsula spatial transfer 2026 Atmosphere

Interpretable Ensemble Machine Learning for Statistical Emulation of Summer $PM_{2.5}$ Change in GEOS-CF over the Arabian Peninsula: How Far Does Spatially Blocked Skill Transfer?

Emulated the 2020-2025 change in summer $PM_{2.5}$ over 3,933 GEOS-CF cells from 13 satellite and reanalysis predictors. Tree ensembles reached spatially blocked CV R2 0.82-0.84 (linear 0.55), but a Random Forest on latitude and longitude alone scored higher (R2 0.91). The predictor model lost all skill once cells within 300 km were excluded, failed on an unseen eastern sector (r 0.11) and did not transfer to other summers. Plain interpolation filled gaps better than any emulator. Split-conformal 90% intervals covered 82-98% of test cells, but only 60% in the east. Limitation: the target is model output, not observations. Relevance: high - a clean demonstration that blocked-CV R2 in PM mapping can be pure interpolation; the same diagnostics belong in sensor-fusion papers.

doi: 10.3390/atmos17101035

Barnett-Itzhaki hourly compound heat air pollution extremes 2026 Atmos Pollut Res

Hourly Resolution Reveals Hidden Temporal Windows of Compound Heat-Air Pollution Extremes Masked by Daily Averaging

METADATA ONLY. Title and author only; no abstract retrievable at harvest. The title claims that hourly data expose compound heat-air-pollution extreme windows that daily means hide. Pollutants, location, thresholds, record length and how many events are missed by daily averaging cannot be stated. Limitation: unverifiable beyond the title; compound-extreme definitions are threshold-dependent, and hourly co-occurrence of heat and O_3 is expected by photochemistry, so the PM-specific contribution is unknown. Relevance: moderate - sub-daily exposure windows are precisely what continuous low-cost monitors resolve and daily regulatory filters do not.

doi: 10.1016/j.apr.2026.103243

Bloise water-soluble carbon nitrogen PM_{10} Chennai street vertical 2026 Atmosphere

Seasonal, Spatial, and Near-Surface Vertical Variability of Water-Soluble Carbon and Nitrogen in PM_{10} in Chennai, India

Three Chennai street environments across three campaigns, sampled at nominal heights of 3, 16 and 37 m: 76 PM_{10} samples, 75 WSOC/WSTN and 74 WSIN determinations. Means: PM_{10} 65.3 $\mu g\,m^{-3}$, WSOC 2.53 ug C/m3, WSTN 1.28 and WSIN 1.26 ug N/m3. WSOC fell from 4.01 to 0.62 ug C/m3 between campaigns while WSIN rose from 1.12 to 1.86. Site contrasts were strong for PM_{10}, and vertical differences were not significant. Limitation: about 25 samples per campaign split across sites and heights gives little power for the vertical null. Campaigns confound season with meteorology. Relevance: medium - no vertical gradient between 3 and 37 m supports rooftop siting of street-canyon LCS for PM_{10}, though the null is underpowered.

doi: 10.3390/atmos17101033

Chen ambient $PM_{2.5}$ representativeness children microenvironments 2026 Build Environ

Evaluating the representativeness of ambient $PM_{2.5}$ for children's exposure and source characteristics across microenvironments during school semesters

METADATA ONLY. No abstract at harvest. The title indicates a comparison of ambient $PM_{2.5}$ with children's exposure and source profiles across school-semester microenvironments. Recorded for coverage; no estimate inferred. Relevance: high in principle - quantifying the ambient-to-personal gap for children is precisely what portable low-cost monitors are deployed for, and the full text should be checked.

doi: 10.1016/j.buildenv.2026.115452

Chen primary brown carbon China 2005-2020 trends radiative effect 2026 Atmos Chem Phys

Long-term evolution and effects of primary brown carbon aerosol in China

CTM simulation of primary BrC across China, 2005-2020, with an internal-mixing optical model. Mean surface PBrC 0.81 ugC/m3; residential, industrial and agricultural combustion contributed 57%, 22% and 18%, together 91% of column PBrC in 2010. PBrC fell 20.8% over the period vs 8.1% for $PM_{2.5}$; direct radiative effect fell from +0.032 to +0.023 W/m2 (-25.7%), 84.2% anthropogenic. Limitation: BrC emission factors and absorption Angstrom exponents are poorly constrained, and evaluation against measured absorption is not described in the abstract. Relevance: medium - multi-wavelength absorption (BrC) channels on sensor nodes could test the residential-combustion dominance.

doi: 10.5194/acp-26-14867-2026

Cheng LETKF WRF-Chem CALIOP component-resolved aerosol assimilation East Asia 2026 Atmos Chem Phys

Constraining the vertical profiles of aerosol component and heating rate over east Asia through assimilating CALIOP observations

Component-resolved LETKF assimilation in WRF-Chem of CALIOP-MODIS synergistic species-specific extinction profiles (dust, sea salt, BC, water-soluble), with spatial and observational constraints. The analysis reconstructs multi-layer vertical structure and corrects severe underestimation of elevated BC plumes, including persistent free-tropospheric BC over South Asia; AERONET and AD-Net lidar validation shows gains in AOT, extinction and single-scattering albedo. Limitation: CALIOP's narrow curtain and 16-day repeat leave most of the domain unobserved, the species classification is retrieval-model dependent, and the abstract gives no quantitative skill figures; surface $PM_{2.5}$ is not evaluated. Relevance: moderate - column and vertical constraints are what satellite-surface fusion with LCS networks currently lacks.

doi: 10.5194/acp-26-14671-2026

Dayanandana CMIP6 dust optical depth MENA evaluation 2026 Atmos Environ

Evaluation of Dust Optical Depth in CMIP6 Simulations over the Middle East-North Africa Region

METADATA ONLY. Title and authors only (Dayanandana, Parottil, Abbas, Al-Harrasi et al.); no abstract at harvest. Evaluates CMIP6 dust optical depth over MENA, presumably against MODIS/MERRA-2 DOD. Models, metrics and biases cannot be stated. Limitation: unverifiable beyond the title; column optical depth maps to surface PM_{10} only through assumed vertical profiles and size distributions, so climate-model DOD skill says little about surface exposure. Relevance: low-moderate - MENA dust is where optical LCS mass error from coarse particles is largest.

doi: 10.1016/j.atmosenv.2026.122431

Ghosh meter-scale heterogeneity new particle formation 3D model 2026 Environ Sci Technol

How Microscale Flow and Vapor Transport Shape New Particle Formation and Growth.

A 10 m resolution 3D multiphysics aerosol model (3D-ICAM) compared with an equivalent box model across emission geometries. Under weak winds (<1 m/s), it reproduced nucleation hotspots, intermittent multi-episode nucleation and persistent bimodal size distributions that the homogeneous case lacks. Resolving spatial structure raised predicted sub-3 nm growth rates up to ~20-fold, bringing them into the observed range, because transport-generated microenvironments relax the growth-scavenging competition. Limitation: idealised geometries; agreement with observations is at the level of growth-rate ranges, not site-specific validation. Relevance: medium - implies point UFP measurements in streets sample sub-grid hotspots; dense number-sensor networks are the natural test.

doi: 10.1021/acs.est.6c10206

He NYC-METS size-resolved organic aerosol offline AMS 2026 Aerosol Sci Technol

Characteristics of size-resolved organic aerosols using offline aerosol mass spectrometry: NYC-METS

METADATA ONLY. Title and authors only; no abstract in Crossref, Europe PMC or Semantic Scholar at harvest. The record reports size-resolved organic aerosol characterised by offline aerosol mass spectrometry - re-aerosolised filter or impactor extracts - within the NYC-METS campaign in New York City. Sampling period, size cuts, source apportionment and sample counts cannot be stated. Limitation: offline AMS loses water-insoluble and semi-volatile fractions in extraction, so size-resolved OA mass is a lower bound; no inference beyond the title is possible. Relevance: moderate - size-resolved organic composition governs refractive index and hygroscopicity, the two terms optical LCS corrections approximate.

doi: 10.1080/02786826.2026.2746516

Imran Chinese cities urban air quality determinants interpretable ML 2014-2023 J Environ Manage

Spatio-temporal variation and determinants of urban air quality in Chinese cities during 2014-2023: an interpretable machine learning analysis

METADATA ONLY. Title and authors only; no abstract retrievable at harvest. The title describes a decade (2014-2023) of spatio-temporal variation in Chinese urban air quality with interpretable ML (SHAP-type attribution) of its determinants. Pollutants, number of cities, predictors and model skill cannot be stated. Limitation: unverifiable beyond the title; SHAP attributions on correlated socio-economic and meteorological predictors are not causal, and the 2014-2023 decline is dominated by the 2013 and 2018 action plans. Relevance: low - monitor-network secondary analysis.

doi: 10.1016/j.jenvman.2026.131184

Kaskaoutis in situ aerosol optical properties eastern Mediterranean 2026 Atmos Environ

In situ aerosol optical properties in the background eastern Mediterranean: seasonal variability, regional transport, and source attribution

METADATA ONLY. Title and authors only (Kaskaoutis, Baalbaki, Bimenyimana et al.). No abstract was retrievable at harvest. The title describes in situ scattering/absorption measurements at a background eastern Mediterranean site, with seasonal variability, regional transport and source attribution. Site, instruments and values cannot be stated. Limitation: unverifiable beyond the title. Relevance: low-medium - background optical properties set the mass-scattering efficiency that optical PM sensors assume.

doi: 10.1016/j.atmosenv.2026.122436

Kim SpaMixed spatial mixed models microbiome $PM_{2.5}$ 2026 Genome Med

Spatial mixed models for assessing environmental exposure effects on the microbiome

SpaMixed is a spatial mixed model for microbiome count data that handles both spatial dependence across regions (conditional autoregressive priors) and ecological correlation among taxa. In simulations it achieved high feature-selection power with low false-positive rates and lower estimation error than existing methods. Applied to two $PM_{2.5}$ studies, it found 'biologically plausible' $PM_{2.5}$-associated taxa. Limitation: the abstract gives no sample sizes, regions or effect sizes for the applications. CAR priors absorb the spatial variance that carries the $PM_{2.5}$ contrast, so the method trades confounding for attenuation. Relevance: low - statistical method, no sensing content.

doi: 10.1186/s13073-026-01770-3

Kirsanov COSMO-ART Saint Petersburg UHI pollution PM_{10} feedback 2026 Atmosphere

Assessing Urban Pollution and Heat Island Feedback in a Northern Megacity Using the Fully Coupled COSMO-ART Model

COSMO-ART with TERRA_URB for winter and summer episodes in Saint Petersburg. UHI-driven mixing and deeper PBL cut near-surface primary pollutants 15-54% (winter) and 5-33% (summer), PM_{10} peaks by >44%, while O_3 rose. Aerosol radiative feedback on the UHI was weak (hundredths of a degree, peaks <0.3 C), cooling in summer and warming over winter snow, under a fifth of canopy forcing. Limitation: two episodes, model-only; no comparison with observed PM_{10} is reported in the abstract, and PM_{10} in a snowy city is dominated by road-dust resuspension that emission inventories handle poorly. Relevance: moderate - urban/rural LCS gradients conflate emission density with UHI-enhanced dilution.

doi: 10.3390/atmos17101030

Langford cooking dominant primary PM_1 eddy covariance London Beijing Delhi 2026 npj Clim Atmos Sci

Traffic and combustion emission controls leave cooking as a dominant local source of primary PM_1 in megacities

Eddy-covariance fluxes with aerosol mass spectrometry and a new source-attribution method in three megacity neighbourhoods. In Delhi, traffic and combustion dominate primary PM_1 emission. In London and Beijing, cooking is the largest local primary source at mealtimes: up to 83% of midday and 87% of early-evening emission in London, 61% and 76% in Beijing. Cooking is absent from national, European and global inventories, though the London Atmospheric Emissions Inventory's estimate matched. Primary emission sources differ from those controlling ambient PM_1 mass, which is secondary-dominated. Limitation: one flux footprint per city and non-refractory PM_1 only (no BC or dust). Relevance: high - source-resolved flux benchmarks for inventory checks; cooking plumes are a known optical-sensor artefact at street level.

doi: 10.1038/s41612-026-01583-6

Li atmospheric cadmium concentration deposition China modelling 2026 Atmos Environ

Modeling the spatial distributions and seasonal variations of atmospheric cadmium concentration and deposition in China

METADATA ONLY. Title and authors only; no abstract retrievable at harvest. National modelling of atmospheric cadmium - carried almost entirely on fine particles - with its spatial and seasonal distribution and deposition. Model, inventory, resolution and evaluation against measured Cd cannot be stated. Limitation: unverifiable beyond the title; metal inventories for smelting and coal are the dominant uncertainty. Relevance: low - mass-only LCS cannot see toxic trace-metal fractions, which is why such composition models remain needed.

doi: 10.1016/j.atmosenv.2026.122427

Liu Beijing autumn-winter $PM_{2.5}$ oxidative potential source predictors 2026 Atmos Environ

Autumn-winter variability and source-related predictors of $PM_{2.5}$ oxidative potential in urban Beijing

METADATA ONLY. Title and authors only; no abstract retrievable at harvest. An autumn-winter campaign relating $PM_{2.5}$ oxidative potential in urban Beijing to source-related predictors (heating, traffic, secondary formation presumably). Assay (DTT, AA, GSH), sample count and predictive model cannot be stated. Limitation: unverifiable beyond the title; a single season and site. Relevance: high - follows the 19 Oct Guangdong OP finding; if OP is predictable from source tracers, a subset of networked sensors with composition proxies could map it.

doi: 10.1016/j.atmosenv.2026.122428

Liu $PM_{2.5}$ sources mixing state Chengdu Kangding altitude 2026 Environ Monit Assess

Altitudinal and seasonal differences in PM(2.5) sources, secondary formation, and particle mixing states in the basin-plateau transition zone of Southwest China.

Summer and winter $PM_{2.5}$ sampling at Chengdu (500 m) and Kangding (2,500 m). Winter $PM_{2.5}$ 79.4 vs 15.2 $\mu g\,m^{-3}$, summer 29.1 vs 8.1. Inorganic ions dominated in Chengdu, carbonaceous species in Kangding; sulfate in summer gave way to nitrate in winter. PMF: Chengdu combustion 26.0%, traffic 32.4%, secondary 35.3%; Kangding combustion 47.9%. Single-particle data showed more SIA particles in Chengdu (29.0% vs 17.6%) and more organic-SIA internal mixing in Kangding. Limitation: two seasons, two sites; the altitude contrast is confounded with city size and heating fuel. Relevance: low - descriptive regional chemistry.

doi: 10.1007/s10661-026-16102-6

Lu street-view greenery street canyon PM concentrations 2026 Environ Sci: Atmos

Association between urban greenery and particulate matter concentrations in street canyons: a local-scale study based on street view imagery

Street-view imagery quantified street greenery quantity and configuration, which were related to PM concentrations in street canyons; the authors attribute the associations to low-level vegetation barriers restricting ventilation. The record's abstract is two sentences: no city, sample size, PM metric, measurement method or effect direction and magnitude are given. Limitation: on what is available, the inference is correlational and the proposed ventilation mechanism is untested; treat as near-metadata until the full text is read. Relevance: moderate - siting of street-level sensors relative to hedges and low canopy is a known representativeness issue for LCS networks, and image-derived greenery is a cheap covariate for spatial calibration.

doi: 10.1039/d6ea00086j

Magalhaes extreme rainfall landslides PM_{10} $PM_{2.5}$ Sao Sebastiao 2026 Environ Monit Assess

Can extreme rainfall associated with landslides affect air quality? A case study in a coastal city (Sao Sebastiao, Sao Paulo State, Brazil)

Hourly PM and meteorology from December 2021 to February 2025, with k-means meteorological clusters and openair polar and time-variation analysis around the February 2023 extreme rainfall and landslides. Rain first scavenged PM. The post-landslide period then showed PM_{10} up 124% and $PM_{2.5}$ up 93.3%. Two comparison rain events showed similar scavenging but different recovery, so the authors do not attribute the rise to landslides alone. Limitation: a single station and one event; clean-up traffic, resuspension of dried debris and meteorology are not separated, and there is no composition data. Relevance: medium - post-disaster resuspension is a case for rapidly deployable sensor nodes.

doi: 10.1007/s10661-026-16071-w

Mai population-weighted $PM_{2.5}$ O_3 China 2015-2024 preprint 2026 Research Square (preprint)

Population Weighted Ambient $PM_{2.5}$ and O_3 Concentrations in China from 2015 to 2024

Ecological geospatial analysis of annual ~1 km CHAP $PM_{2.5}$/O_3 rasters matched to WorldPop counts, 2015-2024. Population-weighted $PM_{2.5}$ fell from 53.06 to 31.25 $\mu g\,m^{-3}$ (-41.1%) while population-weighted O_3 rose from 77.06 to 96.75 $\mu g\,m^{-3}$ (+25.6%); the population-weighted minus geographic gap narrowed for $PM_{2.5}$ (5.77 to 2.43) and widened for O_3 (1.01 to 2.72). 97.4% of 0.2-degree blocks show falling $PM_{2.5}$ with rising O_3 at a 0.5 $\mu g\,m^{-3}$/y threshold. Limitation: purely descriptive and inherits CHAP's ML smoothing; no uncertainty on the weighted means; preprint, not peer reviewed. Relevance: moderate - the narrowing urban-rural $PM_{2.5}$ gap shifts LCS value toward peri-urban and rural siting.

doi: 10.21203/rs.3.rs-10964316/v1

Makacha PRECISE personal ambient $PM_{2.5}$ monitoring framework pregnancy SSA 2026 Front Glob Womens Health

An integrated climate-structured personal and ambient air pollution monitoring framework embedded within a pregnancy and postpartum cohort in Sub-Saharan Africa

PRECISE cohort: 343 pregnant or postpartum women (Gambia 160, Kenya 105, Mozambique 78), two 5-day wearable deployments in opposing seasons recording $PM_{2.5}$, T, RH and GPS/accelerometry; six fixed health-facility monitors. Calibration by repeated pre-deployment co-location with reference instruments and in-country Deming-regression harmonisation. Yield 3,190 person-days personal and 328 ambient days; inter-sensor R2 >0.72 pre-deployment and >0.75 in country; ambient capture >85% in Gambia and Kenya, lower in Mozambique (connectivity). Limitation: inter-sensor R2 is precision, not accuracy - the reference co-location bias and slope, and the RH correction applied, are not in the abstract; 5-day windows in a purposive subsample limit pregnancy-average inference. Relevance: high - a reusable LCS QA protocol for low-resource cohorts.

doi: 10.3389/fgwh.2026.1849628

Mulena dust intrusion regimes north-central Argentina vertical structure optical properties 2026 Atmos Res

Dust intrusion regimes over north-central Argentina: Vertical structure, optical properties, and transport pathways

METADATA ONLY. Metadata only: no abstract at harvest. The title describes a classification of dust-intrusion regimes over north-central Argentina by vertical structure, optical properties and transport pathway, likely from lidar or sun-photometer records with trajectory analysis. No data can be summarised. Limitation: not appraisable; column and vertical optical properties do not by themselves give surface PM. Relevance: low-medium - dust-regime classification is the prior a satellite-surface fusion model needs for coarse-mode episodes in an under-monitored region.

doi: 10.1016/j.atmosres.2026.109404

Nolles IGP vs NCP $PM_{2.5}$ pollution forming efficiency CAMS 2026 Environ Sci Pollut Res Int

Indo-Gangetic Plain experiences higher aerosol pollution than North China Plain despite lower emissions

CAMS EAC4 reanalysis, CAMS-GLOB-ANT emissions and ERA5 for 2019. The NCP emitted 65% more primary PM and precursors (28,860 vs 17,464 kt) yet the IGP had higher annual $PM_{2.5}$ (123.6 vs 113.2 $\mu g\,m^{-3}$); NCP logged more bad-air days (251 vs 238) but the IGP longer episodes, up to 170 consecutive days around Delhi. A 'pollution forming efficiency' was nearly double in the IGP in winter, attributed to winds <6 km/h and PBL <300 m. Limitation: both concentration and emissions are model products for one year; EAC4 assimilates AOD and is biased in the IGP, and emission inventories under-count agricultural and residential burning there, which inflates the efficiency ratio. Relevance: moderate - quantifies what reanalysis cannot replace ground networks for.

doi: 10.1007/s11356-026-38305-7

Park Paris UFP number emissions mass-number conserving 2026 J Hazard Mater

Reconciling mass and number emissions through particle size: Implications for urban ultrafine particle modelling

Summer 2021 simulations from regional background to street scale, with a mass- and number-conserving algorithm converting sectoral PM mass emissions to sectional PN. Scenarios: COPERT vs on-board PN factors for exhaust, non-exhaust traffic, and non-combustion industrial processes including construction. The reference run, using mass emissions plus representative sector diameters, reproduced total PN at background and traffic stations; non-exhaust contributed about a third of traffic PN; the size distribution near 10 nm remained poorly reproduced. Limitation: one summer, and agreement on total PN can hide compensating errors across modes; sub-10 nm emission factors are missing. Relevance: moderate - PN, not $PM_{2.5}$ mass, is the UFP exposure metric, and low-cost condensation counters are the only route to network density.

doi: 10.1016/j.jhazmat.2026.143879

Perez-Vizcaino PM_{10} sources Riotinto mining multi-time-resolution PMF 2026 J Hazard Mater

Characterization of PM(10) sources in the Riotinto copper mining district (SW Spain) using multi-time-resolution receptor modelling.

Year-long (July 2024-July 2025) campaign combining an hourly XRF metals monitor and an Aethalometer with daily high-volume quartz-filter chemistry, merged in multi-time-resolution PMF. Seven sources: waste rock 4.9, regional + traffic 3.6, open pit 1.6, combustion 1.2, aged marine 1.2, dust 1.1 and copper concentrate 0.7 $\mu g\,m^{-3}$. Daily species identified traffic tracers and SOA that the hourly subset alone missed. Arsenic was the main geochemical anomaly. Limitation: one receptor site; multi-resolution PMF weights the daily data heavily and factor uncertainty (bootstrap/DISP) is not reported in the abstract. Relevance: medium - hourly XRF + BC is the reference tier that low-cost networks around mines would need to be anchored to for metal-rich dust.

doi: 10.1016/j.jhazmat.2026.143869

Savolainen road traffic emission factors impact factors Helsinki street canyon 2026 Atmos Chem Phys

From direct emission factors to inverse and indirect impact factors of road traffic: influence of emerging and unregulated pollutants

Wintertime supersite measurements in a central Helsinki street canyon. Fleet emission factors were derived by normalising pollutant increments to CO_2 as a fuel-consumption tracer, across particle number and size, BC, PM_{10}, $PM_{2.5}$, organic and inorganic PM components, NOx and VOCs including PAHs. PM and NOx emission factors were well above the post-2014 Euro limits, so older and heavy-duty vehicles dominate. Traffic showed negative 'impact factors' for O_3 and sub-2 nm small ions and contributed terpenoid VOCs. Limitation: one canyon in one season; the CO_2-ratio method assumes co-emission and common dilution, which fails for secondary and non-exhaust species. Relevance: high - a CO_2-normalised increment is a method LCS nodes with NDIR CO_2 can apply directly.

doi: 10.5194/acp-26-14823-2026

Tripathi satellite-reanalysis sulfate aerosol India ground measurements 2026 Environ Pollut

Improving Satellite-Reanalysis Sulfate Aerosol Estimates over India Using Ground Measurements

METADATA ONLY. Title and authors only (Tripathi, Sahu, Sharma, Gupta); no abstract indexed at harvest. The title indicates ground measurements used to correct satellite-reanalysis (likely MERRA-2/CAMS-type) sulfate aerosol over India. Data sources, method and skill cannot be stated. Limitation: unverifiable beyond the title; Indian speciation networks are sparse and urban-biased, so any correction trained on them risks transferring urban bias to rural cells. Relevance: moderate - component-resolved surfaces are the complement LCS mass networks lack; pairs with the sulfate-dominant AF signal in today's Medicare cohort.

doi: 10.1016/j.envpol.2026.129391

Tsai mineral dust biomass burning aerosol convergence East Asia western North Pacific 2026 Atmos Environ

Convergence of mineral dust and biomass-burning aerosols over East Asia and the western North Pacific

METADATA ONLY. Metadata only: no abstract was available at harvest from any route tried. The title describes convergence of mineral dust and biomass-burning aerosol over East Asia and the western North Pacific, presumably from observation and transport analysis; the instruments, period and metrics cannot be stated. Limitation: not appraisable. Relevance: medium in principle - mixed dust-smoke plumes combine coarse non-spherical and fine absorbing particles, a regime where optical sensor size and refractive-index assumptions fail together.

doi: 10.1016/j.atmosenv.2026.122444

Xiao NCP Sichuan haze reduced-nitrogen NOC oligomerisation $PM_{2.5}$ 2026 Environ Sci Technol

Aqueous Carbonyl-Amino Aggregation Drives Atmospheric Nitrogen-Containing Oligomer Formation in China Haze

Winter-haze $PM_{2.5}$ from the North China Plain compared with the Sichuan Basin. Reduced-nitrogen NOCs were 40% of ESI+ organics. High-molecular-weight (m/z>250) Re-NOCs rose in humid versus dry haze and correlated with NH4+ and the aqueous-SOA fraction of water-soluble organics, alongside longer aliphatic chains and higher unsaturation - read as carbonyl-amino oligomerisation and cyclisation. A marker oligomer ($C_8H_{13}N_3O_2$) tracked WSON; stronger acidity and higher temperature favoured oligomerisation over cyclisation, enriching it in the Sichuan Basin. Limitation: ESI+ is semi-quantitative and ionisation-biased, number of samples is not stated, and mechanism is inferred from correlation. Relevance: low - brown-carbon NOCs add absorbing mass that optical scattering sensors misweight under humid haze.

doi: 10.1021/acs.est.6c09940

Zhao $PM_{2.5}$-O_3 co-pollution episodes BTH YRD 2026 Atmos Pollut Res

Episode analysis of the characteristics and causes of $PM_{2.5}$–O_3 co-pollution in Beijing-Tianjin-Hebei and Yangtze River Delta, China

METADATA ONLY. Title and authors only (Zhao, Xu, Gao, Zhang et al.). No abstract was retrievable at harvest. The title describes an episode analysis of simultaneous $PM_{2.5}$ and O_3 exceedance in BTH and the YRD, including causes. Data sources, period and attribution method cannot be stated. Limitation: unverifiable beyond the title. Relevance: low-medium - co-pollution episodes are where PM-only alerts from sensor networks understate risk.

doi: 10.1016/j.apr.2026.103248

de la Cruz automated forced air injection cordwood stove $PM_{2.5}$ particle size 2026 Atmos Environ

Less Mass of $PM_{2.5}$ Emissions, Smaller Particles: The Trade-Off of Automated Forced Air Injection in a Residential Cordwood Stove

METADATA ONLY. Metadata only: no abstract was available from PubMed, Europe PMC, Crossref or Semantic Scholar at harvest. The title reports that automated forced-air injection in a residential cordwood stove lowers $PM_{2.5}$ mass emissions but shifts particles to smaller sizes. No numbers, test protocol or fuel details can be summarised. Limitation: cannot be appraised from the title; the trade-off claim needs number-based metrics to evaluate. Relevance: high in principle - a mass reduction with a shift to ultrafine sizes is precisely the case where optical mass sensors, blind below ~0.3 um, would overstate the improvement.

doi: 10.1016/j.atmosenv.2026.122443

Cardiovascular & metabolic 12 records

Carried from the daily issues — full summaries (12 of 12 records in this subtopic)

Brugge HEPA sham crossover cytokines visceral adiposity 2026 Am J Prev Cardiol

Pro-inflammatory cytokines moderated by visceral adiposity from a pragmatic randomized crossover trial of portable HEPA air purifiers

116 adults (mean age 41.5) randomised to 30 days HEPA then sham, or the reverse, with a 30-day washout; ultra-sensitive cytokine assays, linear mixed models. No overall IL-1beta difference. Visceral adiposity (ABSI z >=+0.229) moderated the effect (interaction p=0.001): in the 12 participants with excess VAT, HEPA cut mean IL-1beta by 22.8% (0.138 vs 0.179 pg/mL, p<0.001); IL-6 showed a similar non-significant pattern and hsCRP little change; BMI moderated less well than ABSI. Limitation: the positive finding rests on 12 people in a post-hoc-looking subgroup chosen among several adiposity metrics, and achieved PM reduction is not reported in the abstract. Relevance: high - sham-controlled filtration with measured indoor PM is where calibrated low-cost monitors define the delivered dose.

doi: 10.1016/j.ajpc.2026.101775

Du Air Health Index acute myocardial infarction 325 cities 2026 Environ Sci Technol

Establishing an Air Health Index for Predicting the Risk of Acute Myocardial Infarction Onset Associated with Air Pollution and Ambient Temperature

1,207,625 AMI cases across 325 Chinese cities; excess risks from $PM_{2.5}$, NO_2, O_3 and temperature aggregated into a normalised AHI. Per IQR, AHI associated with +5.97% (5.32-6.63%) AMI incidence vs +3.33% (2.93-3.72%) for AQHI and +2.04% (1.73-2.35%) for AQI; near-linear, no threshold; validated on an independent external dataset; 8,534 per 100,000 AMI cases attributed. Limitation: the index is fitted to the same outcome it is scored on, so in-sample superiority over AQI/AQHI is expected; IQR-scaled comparisons across indices with different distributions are not like-for-like, and 'attributable' fractions for a composite that includes temperature mix avoidable and non-avoidable risk. Relevance: high - an outcome-weighted index is a natural output layer for real-time LCS networks, provided the $PM_{2.5}$ input is calibrated.

doi: 10.1021/acs.est.6c07635

Hahad exposome heart failure older adults review 2026 Curr Heart Fail Rep

Exposomic Determinants in Heart Failure Among Older Adults - Mechanistic Links, Clinical Evidence, and Future Perspectives

Narrative review arguing for a life-course, systems-level view of heart failure in older adults in which air pollution sits alongside traffic noise, temperature extremes, metals, artificial light at night and social environment. Mechanisms proposed to converge on oxidative stress, mitochondrial dysfunction, inflammaging, impaired autophagy, epigenetic change and senescence, lowering the threshold for HF onset and decompensation. No new data or pooled estimates. Limitation: the mechanistic convergence is asserted from separate literatures; co-exposures (PM, noise, heat) are spatially correlated, and the review does not address how to separate them. Relevance: low - multi-exposure co-located sensing (PM, noise, temperature) is what the exposome framing would need.

doi: 10.1007/s11897-026-00781-w

Huang $PM_{2.5}$ coronary artery dilatation Kawasaki disease case-crossover 2026 Eur J Pediatr

Association between PM(2.5) and coronary artery dilatation with Kawasaki disease in China

847 children with Kawasaki disease, 2017-2021; case-crossover with a distributed-lag Poisson GLM on coronary Z score. Risk of coronary artery dilatation rose at lags 2-4, peaking at lag 3 with RR 1.11 (1.02-1.21); the abstract does not state the $PM_{2.5}$ increment. Mediation via CRP, ALT and haemoglobin. Limitation: a case-crossover needs an acute event with a datable onset, yet CAD is detected at echocardiography whose timing follows admission and IVIG, so the referent-window logic is shaky; the 'mediators' are markers of disease severity measured after onset, and the highlights' claim of a 'population-based cohort' contradicts the design. Relevance: low - city-monitor assignment, no sensing implication.

doi: 10.1007/s00431-026-07527-9

Lee CRCS-K post-discharge SO_2 stroke recurrence PM null 2026 J Stroke

Post-Discharge Ambient Sulfur Dioxide Exposure and Stroke Recurrence After Acute Ischemic Stroke

27,340 acute ischaemic stroke patients (2014-2021), district-level IDW-interpolated pollutant means over the 3 months after discharge; cause-specific Cox with death as competing risk, cluster-robust SEs, splines. 764 recurrences, 542 deaths over a median 364.8 days. PM_{10} and $PM_{2.5}$ showed no association; only SO_2 did (overall spline p=0.024), with Q4 vs Q1 HR 1.19 (0.94-1.50), stronger in age >=75 and women (interaction p 0.051, 0.062). Limitation: district IDW from sparse monitors gives Berkson-type error that biases PM toward the null, and SO_2 concentrates in harbour cities (Incheon, Ulsan, Busan) - a place effect cannot be excluded. Relevance: moderate - a null PM result built on interpolation is exactly where denser networks change the answer.

doi: 10.5853/jos.2026.03265

Rao $PM_{2.5}$ components cardiorenal multistate trajectory 2026 Int J Cardiol

Associations of $PM_{2.5}$ components with dynamic and accelerated progression of different kidney function states and subsequent cardiovascular events

307,639 participants, 1,058,210 person-years, 23,948 CVD cases and 7,763 CVD deaths; multistate and accelerated failure time models over eGFR categories 1-4. Per IQR, transition from category 2 to 3 had HR 1.273 (1.257-1.289) for sulfate, 1.304 (1.290-1.317) for nitrate and 1.280 (1.267-1.294) for ammonium, larger than for OM or BC. AFT: predicted 2-to-3 transition time 1,727 days at 4.2 vs 869 days at 7.0 $\mu g\,m^{-3}$ nitrate. Limitation: HRs near 1.3 per IQR with CIs this narrow, for highly collinear secondary inorganic components, signal spatial confounding by region more than component-specific toxicity; single-pollutant component models cannot separate them. Relevance: moderate - component attribution needs speciation that optical LCS lack.

doi: 10.1016/j.ijcard.2026.134999

Sun long-term air pollution lipid profiles Shanxi repeated measures 2026 Front Public Health

Long-term exposure to ambient air pollution and lipid profiles: a longitudinal study in a heavily industrialized region of northern China

Repeated-measures study of 9,040 adults with 19,113 lipid measurements from a health-screening programme, 2019-2023. Per 10 $\mu g\,m^{-3}$ $PM_{2.5}$: TG +1.85%, TC +1.70%, LDL-C +1.81%, HDL-C -1.55% (linear mixed models); PM_{10}, NO_2, SO_2 and O_3 in the same direction, CO weaker; age modified the HDL-C association after FDR correction. Limitation: a health-screening population self-selects; CHAP is itself trained on the sparse Shanxi monitor network that underrepresents coal-industry hotspots, and with five correlated pollutants the '$PM_{2.5}$ most pronounced' ranking is not identified from single-pollutant models. Lipids are sensitive to fasting status, not mentioned. Relevance: moderate - industrial-region exposure surfaces are where LCS densification would most change CHAP errors.

doi: 10.3389/fpubh.2026.1950282

Wan $PM_{2.5}$ mortality urban rural elderly Beijing cohort 2026 Zhonghua Liu Xing Bing Xue Za Zhi

Association between fine particulate matter exposure and all-cause and cardiovascular mortality in urban and rural elderly population in Beijing

Beijing Elderly Comprehensive Health Cohort, n=3,472. Per 1 $\mu g\,m^{-3}$ $PM_{2.5}$: all-cause mortality HR 1.053 (1.026-1.080), CVD mortality HR 1.046 (1.001-1.100). Outdoor physical activity modified the association (P-interaction 0.022): HR 1.087 per $\mu g\,m^{-3}$ without outdoor activity vs 1.026 with it. Limitation: a 5% hazard increase per single $\mu g\,m^{-3}$ is roughly an order of magnitude above the pooled long-term estimates. In a city-scale cohort the $PM_{2.5}$ contrast is mostly urban vs rural, so residual confounding by healthcare access and SES is the likely driver. The physical-activity 'offset' is also confounded by baseline health. Relevance: low - single-city satellite exposure.

doi: 10.3760/cma.j.cn112338-20251204-00870

Wongcharoen short-term $PM_{2.5}$ MACE high-risk population nationwide case-crossover 2026 BMJ Open

Associations between short-term PM(2.5) exposure and major cardiovascular events in high-risk population: a nationwide case-crossover study

METADATA ONLY. Title and authors only (Wongcharoen et al.); no abstract indexed at harvest. A nationwide time-stratified case-crossover of short-term $PM_{2.5}$ and major adverse cardiovascular events restricted to a 'high-risk population' (presumably prior CVD, diabetes or hypertension). Country, n, lag structure and effect size cannot be stated. Limitation: unverifiable beyond the title; case-crossover in a pre-selected high-risk group estimates a conditional acute effect that is not comparable with general-population estimates. Relevance: low-moderate - short-lag acute effects are what real-time LCS alerts are meant to act on.

doi: 10.1136/bmjopen-2026-120476

Yadav PM pollution-accelerated cardiovascular aging PACA framework 2026 Cardiovasc Toxicol

Particulate Matter Exposure and Accelerated Cardiovascular Aging: Mechanistic Evidence, Biomarker Integration, and the PACA Framework

A conceptual review that proposes the 'Pollution-Accelerated Cardiovascular Aging' (PACA) framework to separate transient toxicity from a real shift in cardiovascular ageing trajectory. It collects established pathways (mitochondrial and redox dysfunction, epigenetic change, senescence/SASP, proteostasis, endothelial and autonomic dysfunction, the pulmonary-neural-cardiac axis) and adds a Temporal-Evidence Matrix that weighs persistence after exposure stops and change relative to age-referenced comparators. Limitation: hypothesis-generating with no new data or systematic search. Without validated ageing clocks for the vasculature the matrix is hard to operationalise. Relevance: low - longitudinal personal exposure data are the missing input the framework asks for.

doi: 10.1007/s12012-026-10209-5

Zhang wildfire smoke waves long-term $PM_{2.5}$ cardiovascular hospitalisation Medicare 2026 Environ Res Health

Impact of wildfire smoke waves on the association between long-term exposure to fine particulate matter and cardiovascular hospitalizations among older adults in the United States

65.2 million Medicare beneficiaries aged >=65, 2017-2022, ZIP-year quasi-Poisson. Raising two-year $PM_{2.5}$ from 5 to 9 $\mu g\,m^{-3}$ (WHO guideline to US standard) gave overall CVD RR 1.09 (1.09-1.10) in ZIP-years without smoke waves but 1.04 (1.03-1.04) with them (ratio of RRs 1.05, 1.04-1.06); the pattern held below 9 $\mu g\,m^{-3}$ and was stronger in men, urban and western residents. Limitation: the attenuation is what one expects if smoke spikes inflate the two-year mean with a less toxic-per-mass or differently distributed exposure, so this is an exposure-metric artefact as much as a biological finding; ZIP-level assignment and 10 km smoke grids blur the contrast. Relevance: high - long-term means hide episodic peaks that only high-frequency network data resolve.

doi: 10.1088/2752-5309/aeb0c3

Zhu source-specific $PM_{2.5}$ atrial fibrillation Medicare cohort 2026 Environ Health (ACS)

Source-Specific $PM_{2.5}$ Exposure and Atrial Fibrillation Risk in U.S. Older Adults: A Nationwide Cohort Study

26.3 million Medicare beneficiaries, 2000-2019, 171.6 M person-years, 4.44 M incident AF. Per IQR of source-specific $PM_{2.5}$: secondary sulfate HR 1.056 (1.053-1.058), heavy oil/coal combustion 1.048 (1.046-1.050), secondary nitrate 1.032 (1.030-1.034), vehicle 1.029 (1.028-1.030); biomass burning and dust modest, SOA null overall but positive in the West. Limitation: source profiles come from sparse speciation monitors interpolated by inverse distance over 100 km, so source contrasts are largely regional contrasts; sulfate and nitrate fields are smooth and the per-IQR scaling differs by source, so 'sulfate is most toxic' is not a per-microgram statement. Claims-based AF under-ascertains paroxysmal AF. Relevance: high - speciation, not mass, is what optical LCS cannot give; source-resolved risk raises the value of co-located filter or ACSM nodes.

doi: 10.1021/envhealth.5c00638

Neuro / mental health — 4 records

Carried from the daily issues — full summaries (4 of 4 records in this subtopic)

Baird woodsmoke CD4 T cell brain influx saturated fat diet mice 2026 Neurotoxicology

CD4+ T Cell Activation and Peripheral Immune Cell Influx into the Brain Following Wildfire Smoke Exposure is Modulated by a Saturated Fat Diet

Female C57BL/6 mice exposed to filtered air or woodsmoke, with or without a coconut-oil saturated-fat diet, sampled 1, 14 and 28 days after exposure by high-dimensional flow cytometry. Woodsmoke produced a time- and diet-dependent influx of activated CD4 T cells (CD44+, RORgamma-, FOXP3-) into brain, and LFA-1, VCAM-1 and ICAM-1 peaked one day after the last exposure; taurine and glycerolipid metabolism were upregulated. Limitation: 500 $\mu g\,m^{-3}$ is an extreme-episode dose, females only, and effect sizes are not in the abstract. Relevance: low for sensing - a mechanistic anchor for smoke-neuroinflammation epidemiology.

doi: 10.1016/j.neuro.2026.103589

Kalkbrenner traffic air pollution ADHD Denmark critical windows 2026 Environ Res

Traffic-related air pollution links with attention deficit hyperactivity disorder in Denmark: Developmental periods of susceptibility and sex-specific associations

Danish singletons born 1981-2008: 21,860 ADHD cases and a 32,041-person random subcohort. Exposure in each developmental period was mutually adjusted for the other periods to separate windows. The second year of life dominated: EC >1.13 vs <0.49 $\mu g\,m^{-3}$ HR 1.24 (1.12-1.36), not attenuated by $PM_{2.5}$ adjustment; females HR 1.44 (1.21-1.72) vs males 1.18 (1.00-1.40). $PM_{2.5}$ gave the same window and sex pattern. Polygenic scores for ADHD were adjusted. Limitation: period-specific exposures at the same address are highly collinear, and a window identified by mutual adjustment is sensitive to differential measurement error between periods. Diagnosis-based ADHD in females is also subject to referral bias that varies with urbanicity. Relevance: medium - EC, not $PM_{2.5}$ mass, carries the signal, which argues for absorption (BC) channels in dense networks.

doi: 10.1016/j.envres.2026.125900

Wang multi-omics Mendelian randomization $PM_{2.5}$ depression 2026 Research Square (preprint)

Multi-omics Mendelian Randomization Reveals the Causal Association and Mediating Mechanisms of $PM_{2.5}$ with Depression: A Systematic Study Integrating Transcriptomics, Metabolomics, and Radiomics

LD-score regression found genetic correlation between '$PM_{2.5}$ exposure' and MDD (rg 0.334, P 1.9e-11); two-sample MR gave OR 1.27 (1.07-1.51) for MDD. Two-step MR identified five blood genes, six brain-region genes and two blood fatty-acid ratios as mediators; no mediation through CSF metabolites or imaging traits. Limitation: a GWAS of residential $PM_{2.5}$ indexes where people live, so its instruments capture SES, urbanicity and migration; the genetic correlation itself shows horizontal pleiotropy with MDD is likely, which violates the exclusion restriction. The OR has no exposure unit. Preprint. Relevance: low - not an exposure-measurement result.

doi: 10.21203/rs.3.rs-10994413/v1

Zhang CLHLS $PM_{2.5}$ components cognitive disorders lifestyle mediation 2026 J Glob Health

Association between long-term ambient air pollution exposure and cognitive disorders among older adults in China: the mediating role of lifestyle

Chinese Longitudinal Healthy Longevity Study; Cox, Bayesian WQS and mediation models. Per 10 $\mu g\,m^{-3}$, $PM_{2.5}$ HR 1.005 (1.001-1.009), sulphate 1.030 (1.003-1.056), nitrate 1.020 (1.003-1.037), ammonium 1.033 (1.005-1.062), O_3 1.006; mixture OR 1.028 (1.006-1.040) with O_3 carrying the largest WQS weight. Favourable vs unfavourable lifestyle HR 0.488 (0.336-0.710); lifestyle mediated 8.4-9.5% of the pollutant associations. Limitation: an HR of 1.005 per 10 $\mu g\,m^{-3}$ is at the floor of detectability and the abstract gives neither n, case count nor how cognitive disorder was ascertained (MMSE cut-offs in CLHLS are education-sensitive); 'mediation' by lifestyle that is itself a confounder of residential exposure is not identifiable. Relevance: low - gridded component surfaces, no sensor component.

doi: 10.7189/jogh.16.04307

Respiratory & allergic — 11 records

Carried from the daily issues — full summaries (11 of 11 records in this subtopic)

Argersinger air pollution chronic rhinosinusitis review 2026 Curr Opin Otolaryngol Head Neck Surg

The link between air pollution and chronic rhinosinusitis

A short review of recent work on $PM_{2.5}$ and chronic rhinosinusitis. $PM_{2.5}$ is associated with CRS prevalence and severity and with worse disease-specific and quality-of-life outcomes after endoscopic sinus surgery. In nasal epithelial cells from CRS patients it worsens remodelling and barrier dysfunction and reduces ciliogenesis. Traffic, industrial, wildfire and pollen-related PM contribute independently. Limitation: non-systematic, no pooled estimates. Most cited associations are cross-sectional or claims-based with modelled residential exposure, and CRS phenotype (with or without polyps) is rarely separated. Relevance: low.

doi: 10.1097/moo.0000000000001164

Gonehgar sand dust storms asthma Asia systematic review 2026 J Asthma

Geographic Patterns of Asthma Morbidity: A Systematic Review of Sand and Dust Storm (SDS) Impacts in Asia

Thirty-three studies to May 2023 (13 cohort, 12 panel), mostly Japan, Korea and the Middle East. Nearly all reported positive associations between SDS days and asthma exacerbations, admissions or ED visits, peaking at 1-5 day lags, with larger effects in children and older adults and stronger effects in East Asia than the Middle East. No pooled estimate. Limitation: the search is 29 months old, English-only, and SDS exposure is defined inconsistently (event days vs PM_{10} thresholds), which precludes meta-analysis and may explain the East-West contrast. Relevance: moderate - coarse-mode dust is where optical LCS under-read most, so SDS health evidence built on dust-flagged sensor data would need a coarse-fraction correction.

doi: 10.1080/02770903.2026.2751845

Guerra-Ruiz cytokines volcanic emissions Cumbre Vieja ASHES 2026 Ecotoxicol Environ Saf

Systemic cytokine signatures associated with volcanic emisions exposure and respiratory disease after the cumbre vieja eruption-ashes project.

Cross-sectional ASHES sub-study after the 2021 Tajogaite eruption: 102 adults (66 general population, 13 occupationally exposed, 23 asthmatic). Workers had the longest exposure (median 10 h vs 6 h) and higher IL-2, IL-7, IL-17A and IL-17F; asthmatics showed broader TH1/TH2/TH17 activation, and IL-4 correlated with exposure time in asthmatics (r 0.79). Limitation: exposure is hours of presence, not measured PM or SO_2, and 13 workers is too few for a stable comparison; 48 markers tested without stated correction. Relevance: medium - the eruption had dense monitoring; linking those PM records to individuals would turn this into an exposure-response study.

doi: 10.1016/j.ecoenv.2026.120950

Harb idiopathic subglottic stenosis geographic clustering ecological 2026 Laryngoscope Investig Otolaryngol

Geographic Clustering of Idiopathic Subglottic Stenosis in North America: An Ecological Cohort Study

4,450 members of an online iSGS support group (99.3% female) mapped to state/province and scaled by census White female populations. Prevalence clustered in dry, high, warm areas: Wyoming 18.1, Utah 17.4, Colorado 13.9 per million vs New Hampshire 1.2; Alberta 85.4 vs Quebec 2.9. $PM_{2.5}$ was not independently associated with prevalence but modified the elevation association. Limitation: the numerator is self-selected membership of an online group, so 'prevalence' reflects diagnosis, awareness and referral patterns; state-level ecology cannot separate climate from health-system access. Level 4 evidence by the authors' own grading. Relevance: low - a null for PM at state scale, hypothesis-generating only.

doi: 10.1002/lio2.70607

Jung peaker power plant emissions ED urgent care visits Massachusetts 2026 J Clim Change Health

Individual-level health effects of living near a gas-fired "Peaker" power plant

Primary-care patients living within 30 km of a gas-fired peaker plant, 2017-2021; 2,310 ED/UC visits analysed with a case-crossover design. Exposure was a daily HyADS parcel-trajectory index (mean 1.4, SD 16, reflecting plant intermittency); models adjusted for heat index and non-peaker $PM_{2.5}$ (mean 8.5 $\mu g\,m^{-3}$). Respiratory visits OR 1.044 (1.001-1.090) per unit; non-white patients OR 1.347 (1.034-1.755). Cardiovascular, headache and pregnancy visits were null. Limitation: the exposure is unitless and ZIP-code assigned, so the OR has no concentration scale. Peaker 'on' days coincide with peak heat and grid stress. A lower CI of 1.001 across four outcome families invites multiplicity concerns. Relevance: high - intermittent point sources are exactly what fence-line LCS can time-resolve.

doi: 10.1016/j.joclim.2026.100762

Kim interstitial lung disease changing climate Personal View 2026 Lancet Respir Med

Interstitial lung diseases in a changing climate: emerging risks and opportunities to improve care

A Lancet Respiratory Medicine Personal View on climate-related exposures in interstitial lung disease. Air pollution is the established risk factor. Extreme heat, viral illness, damp housing, pollen, disasters and desertification are framed as emerging threats. Wildfire smoke, occupational ILD under climate change and disaster preparedness are named as the main gaps. Limitation: an opinion synthesis with no quantitative evidence grading. ILD cohorts are small and heterogeneous, so most climate-ILD links are extrapolated from COPD and asthma. Relevance: low-medium - wildfire-smoke alerting for ILD patients is a concrete use case for hyperlocal sensing.

doi: 10.1016/s2213-2600(26)00252-3

Li size-specific PM lung function China Pulmonary Health Study 2026 Environ Health (ACS)

Associations between Size-Specific Particulate Matter and Lung Function: Results from the China Pulmonary Health Study

50,991 adults >=20 y from the China Pulmonary Health Study; multivariable linear models on long-term size-resolved PM. Per IQR PM_1: FEV1 -36.84 mL (-45.91 to -27.76), FVC -34.63 mL (-45.48 to -23.79); large-airway decrements grew as particle size fell, whereas the coarse fraction $PM_{2.5}$-10 gave the largest small-airway (FEF75, FEF25-75) decrements. Limitation: size fractions are derived from correlated modelled surfaces (PM_1 and $PM_{2.5}$ share most of their mass), and the abstract reports no multi-fraction model, so 'size-specific' ranking may reflect differential exposure-model error rather than biology; cross-sectional. Relevance: high - optical LCS report PM_1 and coarse bins natively, and this is evidence that those channels carry distinct health signal worth calibrating, not discarding.

doi: 10.1021/envhealth.5c00676

Meng traffic noise air pollution lung function randomised crossover 2026 Environ Res

Independent and interactive effects of traffic-related noise and air pollution on lung function: A randomized crossover study in healthy young adults.

Randomised crossover with four exposure conditions (noise and air pollution separated by intervention) in 51 healthy young adults beside a busy road at evening rush hour; pollutants and noise measured each session, mixed models adjusted for age, sex and BMI. Per 10 $\mu g\,m^{-3}$ PM_{10}, FEF75 fell 0.215 L/s; per 10 $\mu g\,m^{-3}$ O_3, 0.224 L/s. Noise independently reduced dynamic ventilatory indices, and exploratory interaction terms suggested noise amplified PM_1- and $PM_{2.5}$-associated FEV1 and small-airway decrements. No CIs in the abstract. Limitation: the four interventions are not described in the abstract and exposure conditions cannot be blinded, and pollutant contrasts within a single rush-hour session are small and collinear, so the interaction findings are hypothesis-generating. Relevance: medium - co-located acoustic and PM nodes are needed to separate the two stressors.

doi: 10.1016/j.envres.2026.125963

Morales El Paso port-of-entry $PM_{2.5}$ immigration COPD census tracts 2026 J Immigr Minor Health

Breathing at the Border: $PM_{2.5}$ Exposure, Immigration, and COPD Across Census Tracts Close to Ports-of-Entry

El Paso census tracts; NASA satellite-derived $PM_{2.5}$ merged with CDC tract COPD prevalence and ACS social vulnerability; GLMs with distance to the port-of-entry. $PM_{2.5}$ and percent foreign-born non-citizens were positively associated with COPD prevalence; tracts within three miles of a port had higher prevalence, and the $PM_{2.5}$ effect was stratified by proximity. No coefficients or CIs in the abstract. Limitation: CDC PLACES prevalence is itself small-area modelled from BRFSS demographics, so regressing it on demographic covariates is partly circular; smoking is not addressed; satellite $PM_{2.5}$ at ~1 km cannot resolve truck-idling gradients at bridge queues. Relevance: high - port-of-entry diesel queues are a textbook case for dense LCS siting.

doi: 10.1007/s10903-026-02022-7

Tullis cystic fibrosis CRS olfaction pollutants 2026 Int Forum Allergy Rhinol

Environmental Exposures, Chronic Sinusitis, and Olfaction in People With Cystic Fibrosis

80 adults with CF and chronic rhinosinusitis enrolled 2019-2023 at US rhinology centres. Ozone correlated with worse SNOT-22 (Rs 0.34, p=0.002), QOD-NS (Rs 0.30) and smell identification (Rs -0.30); $PM_{2.5}$, PM_{10}, CO, SO_2 and NO_2 showed no significant associations. Limitation: unadjusted Spearman correlations across six pollutants and several outcomes in 80 people, with exposure from the nearest regulatory monitor to a ZIP centroid - the PM null is uninformative rather than reassuring. Relevance: moderate - a small clinical cohort is the case where residence-level sensor data would most reduce exposure misclassification.

doi: 10.1002/alr.70204

Wang Guangdong $PM_{2.5}$ oxidative potential DTT childhood allergy 2026 Environ Sci Technol

Associations of $PM_{2.5}$ and Its Oxidative Potential with Childhood Allergy-Related Symptoms: A Cross-Sectional Study in Guangdong Province, China

131,346 children aged 6-18 in six Guangdong cities. $PM_{2.5}$ mass and DTT oxidative potential were positively associated with all three symptom outcomes; mass-normalised OP (DTTm) gave the most consistent associations, lower AIC/BIC and the greatest stability in two-pollutant models. Mixture analyses implicated several constituents; OP plausibly carried part of the Co, OC, Cu and Hg associations, the largest indirect share 36.2% for OC and rhinoconjunctivitis via DTTm. Limitation: cross-sectional symptom questionnaires; OP is presumably modelled or assigned from few filter sites, and the abstract reports no effect sizes, so the model-fit comparison cannot be weighed against exposure-error differences between metrics. Relevance: high - OP is not measurable by optical LCS, which bounds what a mass-only network can say about allergy.

doi: 10.1021/acs.est.6c05462

Reproductive & developmental — 6 records

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (6 OF 6 RECORDS IN THIS SUBTOPIC)

Atkin air quality excess heat pregnancy outcomes review 2026 Nurs Clin North Am

Impact of Poor Air Quality and Excess Heat on Pregnancy Outcomes.

Narrative review on combined poor air quality and excess heat in pregnancy. Argues that physiological adaptations of pregnancy raise susceptibility, that structural determinants concentrate exposure in some populations, and that nurses should educate patients, promote adaptation and advocate for climate mitigation. No quantitative synthesis in the abstract. Limitation: non-systematic; heat-pollution interaction is asserted rather than quantified. Relevance: low - clinical guidance context only.

doi: 10.1016/j.cnur.2026.06.010

Falcon-Gracia BiSC placental sncRNA air pollution maternal smoking 2026 medRxiv (preprint)

Impact of air pollution and tobacco smoke on the placental small non-coding RNA landscape

390 chorionic villus samples from the Barcelona Life Study Cohort, small-RNA sequencing; 979 sncRNA clusters (miRNA 33%, piRNA 26%, snoRNA 12%, tRF 8%). Maternal smoking was associated with miR-421 down-regulation and up-regulation of SNORD113/114 snoRNAs at the imprinted DLK1-DIO$_3$ locus and piR-26394. No sncRNA was associated with PM$_{2.5}$, BC or NO$_2$ after multiple-testing correction; several miRNAs were suggestive. Limitation: preprint; a null after FDR across ~1000 features at n=390 says little about effect absence, and pregnancy-average LUR exposure is a blunt contrast for trimester-specific placental biology. Relevance: low - a usefully reported null for PM against a positive smoking control.

doi: 10.64898/2026.10.16.26365732

Ge preconception folic acid air pollution birthweight 2026 Environ Health (ACS)

Preconception Folic Acid Supplementation Associated with Attenuated Effects of Air Pollution on Birthweight Outcomes

METADATA ONLY. Title and authors only (Ge, Zhang, Liao ... Berhane, Bai et al.); no abstract retrievable from PubMed, Europe PMC, Crossref or Semantic Scholar at harvest. The title claims effect modification of the air pollution-birthweight association by preconception folic acid use. Pollutants, cohort, n and effect sizes cannot be stated. Limitation: unverifiable beyond the title; supplement use is self-reported and strongly socially patterned, so 'attenuation' by folate is exposed to confounding by SES and antenatal-care quality unless handled explicitly. Relevance: low - no sensing content evident.

doi: 10.1021/envhealth.5c00779

Kalam HAPIN LPG intervention child growth stunting ages 2-5 2026 medRxiv (preprint)

Effects of a 500-day LPG cookstove and fuel intervention delivered during gestation and infancy on height-for-age (HAZ) and stunting at ages 2, 3, 4 and 5 years

HAPIN randomised trial (~3,200 pregnant women; LPG stove plus free fuel to infant age 12 months), extended follow-up of 2,169 children. The intervention cut gestational PM$_{2.5}$ from 117.5 to 80.7 μg m^{-3} and infant PM$_{2.5}$ from 113.2 to 36.9 μg m^{-3}, but post-trial exposures converged (99.9 vs 98.7). ITT effect on repeated HAZ: beta -0.06 (-0.16 to 0.03); no effect on stunting at any age and no effect modification. Limitation: even the intervention arm stayed far above the WHO guideline during gestation, so this is a null for a partial exposure contrast, not for clean air. The exposure-response analysis that would test dose is still pending. Preprint, not peer reviewed. Relevance: medium - the trial's personal monitoring shows how far stove switching moves exposure.

doi: 10.64898/2026.10.16.26365712

Liu PM$_{2.5}$ components small for gestational age IVF Guangzhou 2026 BMC Public Health

Maternal exposure to PM(2.5) and its components and risk of small for gestational age in women undergoing in vitro fertilization: a propensity score-matched case-control study in Guangzhou, China.

Propensity-score-matched case-control within 1,459 IVF conceptions in Guangzhou: 166 SGA cases matched 1:1. Whole-pregnancy PM$_{2.5}$ OR 1.124 (1.015-1.245); second trimester 1.082 (1.005-1.165); third trimester 1.075 (1.002-1.154); the exposure increment is not given in the abstract. SO4, OM and BC were positively associated; mixture models ranked nitrate as the dominant driver, and no subgroup reached significance. Limitation: 332 matched women give lower CI bounds within 0.2-1.5% of the null across many component-by-window tests, with no multiplicity correction reported; the component estimates come from a modelled product whose component-specific error is unknown. Relevance: low-medium - component attribution here rests on modelled speciation that sensor networks cannot supply.

doi: 10.1186/s12889-026-28800-4

Moradnia Isfahan pregnant urinary metals indoor dust AQI 2026 Environ Geochem Health

Spatial variability of heavy metals in pregnant women: linking urinary biomarkers, indoor dust contamination, and air quality in Isfahan, Iran

80 pregnant women; paired urine and vacuum dust analysed by ICP-OES. Dust metals ranked Pb>Cr>Ni>Cd (Pb mean 111.10 ug/g); urine Pb 10.52, Cr 8.30, Ni 4.14 ug/L. Urinary metals and AQI differed across urban zones ($p<0.001$); dust did not (Pb $p=0.514$). Building age correlated with dust Pb (rho 0.788). Dust and urinary profiles were uncorrelated. Limitation: n=80, spot urine without creatinine adjustment stated, and the only airborne metric is a zone-level AQI - no PM mass or PM-bound metal was measured, so 'air quality' enters only as an ecological covariate. Relevance: low - illustrates the gap that personal or indoor PM monitoring with filter speciation would fill.

doi: 10.1007/s10653-026-03508-x

Mechanistic toxicology 10 records

Carried from the daily issues — full summaries (10 of 10 records in this subtopic)

Amato mobile air-liquid interface toxicity Barcelona harbor urban 2026 Environ Sci Pollut Res Int

Assessment of air pollution toxicity in urban environments in Barcelona, Spain, using a mobile air-liquid interface system

A mobile air-liquid interface system exposed A549 cells directly to ambient air at a harbour site and an urban background site, alongside co-located particle and gas monitoring. Partial least squares explained up to 80% of the variance in ALI toxicity from air-quality parameters. Gas-phase pollutants, ozone above all, contributed more than particles. Limitation: LDH release in a single tumour-derived cell line is a crude endpoint. With two sites and highly collinear predictors, PLS loadings cannot separate ozone from photochemical-age covariates. n and exposure durations are not given in the abstract. Relevance: medium - pairing in-situ bioassays with multipollutant monitors is the template for testing whether mass-based LCS metrics track toxicity.

doi: 10.1007/s11356-026-38306-6

Jiang carbon nanoparticle NSCLC progression ARNT2 ARL14 2026 Ecotoxicol Environ Saf

Carbon nanoparticle exposure promotes the progression of non-small cell lung cancer via the ARNT2/ARL14 axis: Findings from in vivo and in vitro models

Mouse CNP-induced fibrosis model (200 ug/kg every other day, 90 days) identified ARNT2 by transcriptomics. In xenografts with continued CNP exposure, tumour volume at day 36 rose (A549 194 to 308 mm3; A427 245 to 417 mm3) along with Ki-67/PCNA and N-cadherin/vimentin. ARNT2 knockdown reversed these changes. In vitro, 2 ug/mL for 48 h increased proliferation, migration (up to 3.6-fold), invasion, stemness markers and IL-6 (up to 13.7-fold), and reduced apoptosis. Limitation: carbon black nanoparticles stand in for 'the shared core' of $PM_{2.5}$, coal dust and smoke, but lack their organic and metal fractions. Instillation doses and in-vitro concentrations far exceed lung deposition from ambient air. Group sizes are not reported. Relevance: low.

doi: 10.1016/j.ecoenv.2026.120951

Kou Xing Su San Sang Xing Tang $PM_{2.5}$ lung injury rats 2026 Pak J Pharm Sci

Xing Su San and Sang Xing Tang attenuate $PM_{2.5}$-induced lung injury and inflammatory marker expression in rats

40 male Wistar rats in five groups of 8. $PM_{2.5}$ at 500 $\mu g\,m^{-3}$ raised lung RAGE mRNA 2.8-fold (warm-dry) and 2.5-fold (cool-dry), NF-kB p65 2.4- and 2.2-fold, and S100A9-positive cells to 3.45% and 3.13% vs 0.82% in controls (all $P<0.05$). The two herbal formulas, given by gavage, partly reversed these markers in their matched climate model (e.g. RAGE 1.6-fold with Sang Xing Tang). Limitation: 500 $\mu g\,m^{-3}$ is an episode-level dose, n=8 per group, no histology scoring or blinding reported, and formula composition is not standardised. Relevance: low - a temperature-humidity x PM interaction in toxicology, not monitoring.

doi: 10.36721/pjps/2026.40.15845.1

Liu $PM_{2.5}$ IL-17 microglia neuroinflammation 2026 Environ Health (ACS)

IL-17 Signaling Pathway: A Potential Signaling Pathway Mediating Neuroinflammation Induced by $PM_{2.5}$

BV2 microglia exposed to $PM_{2.5}$ lost viability and raised inflammatory cytokines; microarray DEGs enriched for inflammatory GO terms and the KEGG IL-17 pathway with downstream NF-kB and MAPK. An anti-IL-17A antibody reversed cytokine release, microglial activation and behavioural deficits (implying an in vivo arm). Limitation: the abstract gives no $PM_{2.5}$ source, composition or dose, so environmental relevance cannot be judged; BV2 is an immortalised murine line with an exaggerated inflammatory phenotype, and IL-17 is usually T-cell/gamma-delta derived, so microglial IL-17 signalling needs cell-type attribution. Relevance: low - mechanistic only.

doi: 10.1021/envhealth.6c00080

Ruan gut microbiota dose-time trajectories $PM_{2.5}$ PFOS mouse 2026 Environ Pollut

Time-resolved profiling reveals nonlinear dose-time trajectories of gut microbiota in $PM_{2.5}$ and PFOS exposure models

Two mouse models ($PM_{2.5}$ inhalation; PFOS in drinking water) sampled repeatedly across dose levels and exposure stages, with 16S sequencing and generalised additive models for dose-time structure. Controls showed clear temporal succession, which the authors stress must be modelled before attributing change to exposure. In the $PM_{2.5}$ model, restructuring peaked at early-to-intermediate stages, with nonlinear trajectories in Lactobacillus, Prevotella and the Firmicutes/Bacteroidetes ratio; PFOS effects were later and longer. No effect sizes with CIs are reported in the abstract. Limitation: functional pathways are inferred, not measured; doses and particle source are not given in the abstract, and cage effects dominate small murine microbiome studies. Relevance: low - mechanistic, but the time-varying response argues against single-timepoint toxicology.

doi: 10.1016/j.envpol.2026.129383

Sun respirable mining dust natural killer cell dysfunction 2026 Environ Sci Technol

Respirable Particulate Matter Induces Natural Killer Cell Dysfunction via Oxidative Stress-Mediated Lysosomal and Mitochondrial Injury: Implications for Mineral-Rich Mining Dust Exposure.

In-vitro exposure of NK cells to respirable mineral dusts with a K562 target-cell challenge. Injury was dose- and composition-dependent: apoptosis/necrosis, shifted cytokines, altered perforin/granzyme B function and impaired killing. Silica- and silicate-rich dusts were worst; iron-bearing and carbonaceous dusts produced distinct oxidative patterns. Mitochondrial depolarisation, lower mitochondrial mass and lysosomal deacidification converged across dusts; ascorbic acid partially restored organelle function and IFN-gamma. Doses and their relation to inhaled lung burden are not given in the abstract. Limitation: an isolated cell line in submerged culture, no lung-burden dosimetry, so the proposed adverse outcome pathway to lung cancer is inferential. Relevance: low - composition (silica fraction) matters, which mass sensors cannot resolve. Note: this DOI was held from the 3 Oct issue as unauthored and is now carried here.

doi: 10.1021/acs.est.6c07515

Tang CPS-II plasma proteome air pollution lung cancer ever-smokers 2026 Environ Sci Technol

Analysis of the Potential Mediating Role of Plasma Proteins in the Association Between Air Pollution Exposure and Lung Cancer Risk Among Ever-Smokers

95 matched case-control pairs of ever-smokers with pre-diagnostic plasma proteomics from the CPS-II Nutrition Cohort. Seventeen proteins associated with both exposures and lung cancer (CX3CL1, ADAM15, CXCL17 among them, $p<0.05$); high-dimensional mediation (HIMA) flagged six, including CX3CL1 and CXCL17 (Bonferroni $p<0.05$). Limitation: 190 people against a high-dimensional proteome, and the mediation signals disappeared when cases diagnosed within a year of blood draw were excluded - consistent with reverse causation by subclinical tumour. The abstract does not name the pollutants' exposure model. Pathways (chemokine signalling, immune modulation) are hypothesis-generating only. Relevance: low - biomarker discovery, no sensing implication.

doi: 10.1021/acs.est.6c07407

Wang seasonal PM size constituents oxylipin panel Beijing 2026 Environ Sci Technol

Winter and Summer Differences in Oxylipin Responses to Ambient Particles: Roles of Sizes and Constituents

Panel of 123 adults in Beijing with repeated visits in winter and summer; 27 $PM_{2.5}$ mass, size and constituent metrics against five lipoxygenase-derived oxylipins. Associations peaked at lag 0-1 d; across metrics, oxylipin increases ranged 3.3-114.2%, most robust for UFP, accumulation-mode number, OC and chloride. UFP showed an exposure-response only in summer: per IQR, +12.9 to +97.7% in summer vs -7.7 to +18.8% in winter; accumulation-mode and OC effects were larger in winter. No CIs in the abstract. Limitation: 27 metrics x 5 biomarkers x 2 seasons x several lags is a large test space, and the season contrast also changes time outdoors and ventilation. Relevance: high - size-resolved number, not mass, carries the summer signal, which mass-only optical sensors cannot report.

doi: 10.1021/acs.est.6c08647

Zendi brown carbon wood pyrolysis tracers cytotoxicity A549 2026 Chem Res Toxicol

Fine Brown Carbon Aerosol Emitted by Wood Pyrolysis - New Harmful Tracers and Cytotoxicity in Human Lung Cells

Smouldering-regime BrC was generated by 350 C wood pyrolysis, extracted in three solvents and characterised by UHPLC-ESI-ToF-MS non-target analysis (776-1,132 molecular features per extract). Water extracts were dominated by lignin phenols and aromatic acids; methanol and acetone pulled less oxidised terpenoid and sterol-like species. Cytotoxicity in A549 cells ranked water < methanol < acetone, consistent with lipophilic, membrane-partitioning and redox-active compounds; named contributors to baseline narcosis included 3-hydroxybenzoic, salicylic and p-coumaric acids, benzene-1,2,4-triol and coniferylaldehyde. Limitation: solvent extracts at unreported dose-equivalence to inhaled mass, a single carcinoma line, and pyrolysis rather than flaming or aged smoke. Relevance: low-medium - supports BrC (UV-absorbing) channels as a toxicity-weighted metric.

doi: 10.1021/acs.chemrestox.6c00273

de Luna wildfire PM toxicity systematic review 2026 J Hazard Mater

Systematic review on the toxicity of wildfire emitted particulate matter: State of science, knowledge gaps and way forward

46 in vitro and in vivo studies, 2005-2025 (epidemiology excluded). Only 42% of in vitro and 27% of in vivo studies used PM actually collected during wildfires; the rest used surrogate burns. In vitro work concentrates on lung (50%) and macrophages (17%); inhalation studies on lung (46%), heart (15%) and brain (15%). Mechanisms reported are mainly inflammation, oxidative stress and DNA damage, and toxicity varied with dose, size, fuel and burn condition. Gaps: incomplete physicochemical characterisation, unrealistic doses, few chronic or toxicokinetic studies, no work on mixtures with urban pollution. Limitation: no quantitative synthesis or risk-of-bias grading, so heterogeneity is described, not explained. Relevance: medium - the field cannot yet say which smoke property a sensor should track beyond mass.

doi: 10.1016/j.jhazmat.2026.143985

Occupational & indoor — 11 records

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (11 OF 11 RECORDS IN THIS SUBTOPIC)

Alquwayi respirable crystalline silica occupational exposure Saudi Arabia review 2026 Front Public Health

Potential risk of occupational exposure to respirable crystalline silica in Saudi Arabia: a comprehensive review of the current status and future challenges

Single-author review integrating literature, labour and mining statistics, mineralogy and regulation through a 'silica hazard triangle' (workforce, materials, activities). Concludes that mining, construction, cement, glass, engineered stone and photovoltaic manufacturing expansion plus arid-climate dust raise RCS exposure potential, while measured exposure, epidemiology and surveillance data are scarce. Limitation: no new measurements and no quantitative exposure estimates - the conclusion is an inference from sector size, not from data. Relevance: low-moderate - optical LCS cannot speciate silica, so RCS monitoring needs gravimetric/FTIR sampling regardless.

doi: 10.3389/fpubh.2026.1918362

Arif Agra leather footwear facilities indoor $PM_{2.5}$ CO_2 formaldehyde 2026 Research Square (preprint)

Indoor Air Quality and Thermal Environment Across Three Leather Footwear Facilities in Agra India

Three leather-footwear facilities (cottage, small, organised), 27 months, 608 sessions. Facility dominated variance for every parameter (partial eta2 0.10-0.87); cottage vs organised means $PM_{2.5}$ 2.0-fold, CO_2 1.9-, formaldehyde 3.0- and VOC 3.5-fold higher. Season dominated PM and thermal variables, tracking ventilation; indoor/outdoor $PM_{2.5}$ ratios 0.48-1.00 across facilities; CO_2 above the ASHRAE threshold in 99.5% of cottage sessions vs 0% organised. Limitation: preprint; one facility per tier, so scale is confounded with building; instruments and their calibration are not stated, and spot sessions are not shift-weighted personal exposure. Relevance: moderate - informal-sector workplaces are where portable LCS are the only feasible monitor.

doi: 10.21203/rs.3.rs-10962722/v1

Du classroom CO_2 proxy PM VOC ventilation types 2026 Indoor Environments

CO_2 as a proxy for PM and VOCs in classrooms across ventilation types and activities

METADATA ONLY. Title and authors only; no abstract retrievable at harvest. The title tests whether CO_2 tracks PM and VOCs in classrooms across ventilation types and activities - i.e. whether a cheap NDIR sensor can stand in for particle and VOC monitoring. Number of classrooms, ventilation categories, PM instrumentation and agreement statistics cannot be stated. Limitation: unverifiable beyond the title; physically, CO_2 tracks occupant-generated bioeffluents and air exchange, whereas PM indoors also depends on outdoor infiltration, resuspension and filtration, so decoupling under mechanical ventilation is expected. Relevance: high - directly asks whether school CO_2 monitor rollouts can be read as PM exposure surveillance.

doi: 10.1016/j.indenv.2026.100200

Kim pig building air cleaner TSP $PM_{2.5}$ productivity 2026 J Anim Sci Technol

Association between indoor air pollutants and pig productivity index by on-site application of air cleaner

One commercial pig building, control vs air-cleaner-treated room, monitored six months across rearing stages. Air-cleaner reduction efficiency rose with stage, 5-10% at farrowing to 30-45% at fattening; treated rooms showed better weight gain, feed conversion, activity index and mortality. Weight gain and activity correlated with TSP and $PM_{2.5}$ (and airborne fungi). Limitation: one room per arm, no randomisation or replication - room effects are inseparable from treatment; the cleaner technology and PM instrumentation are not stated. Relevance: low - livestock housing is a large agricultural PM source and an occupational exposure for farm workers, but this design supports neither inference.

doi: 10.5187/jast.2500087

Kuye ultrafine particle respiratory deposition dose wood stoves UK homes 2026 Atmosphere

Respiratory Deposition Dose of Ultrafine Particles from Improved Wood Stoves and Open Fireplaces

Exploratory measurements in four UK homes (Ecodesign stove, multifuel Ecodesign stove x2, open fireplace), 11-12 winter days each, 1-min PNC combined with age/sex/activity ventilation rates and the ICRP model. Daily inhaled dose 0.45-16.0 x10^9 particles/kg/day sitting and 2.37-73.2 x10^9 at moderate activity (4.5-5.3-fold higher). Child scenarios had 2.7-3.8x the adult body-mass-normalised dose; household mean deposition rate 1.50-5.99 x10^8 particles/min. Limitation: n=4 homes that differ simultaneously in appliance, fuel, room, ventilation and behaviour, so stove-type effects cannot be identified; doses are modelled from concentration, not measured. Relevance: medium - residential UFP is invisible to mass sensors; indoor PNC nodes are the relevant instrument.

doi: 10.3390/atmos17111039

Matuskova wood stove phase-resolved particle-bound PAHs 2026 Environ Res

Characterisation of particle-bound PAHs emitted from a residential wood-burning stove and their dependence on combustion phases

Three temperate wood species burned in a standard residential stove, aerosol split by cascade impactor across five combustion phases. PM peaked at ignition and flaming; PAHs followed for spruce and black locust, while beech peaked later (interval 4), coinciding with low flue-gas oxygen. Size distributions were monomodal at ~0.1 um aerodynamic diameter with PM_1 70-89% of mass. Hardwoods emitted mainly LMW PAHs; spruce gave an initial HMW burst, shifting to LMW during burnout, when PAHs moved towards coarser particles. Limitation: one stove under nominal operation; real-world overloading and damp fuel dominate emissions. Relevance: moderate - an ~0.1 um mode sits below optical LCS detection efficiency, so wood-smoke PM is under-counted by size-blind calibrations.

doi: 10.1016/j.envres.2026.125931

Mohammed mobile HEPA-UV air decontamination operating theatre 2026 Infect Prev Pract

Evaluation of a mobile surgical air decontamination device in reducing airborne microbial load in the operating theatre of a tertiary care hospital.

Before-after evaluation of a mobile HEPA + UV decontamination unit in one operating theatre over 15 days (July 2025). Mean total particle count fell from 6,836 to 4,072 particles/m3 (-40.4%), significant in the 0.3 and 0.5 um channels; CFU fell from 34.5 to 15.2 CFU/m3 (P 0.003). Limitation: one theatre, short before-after design with no control period or blinding to activity level, and no clinical outcome (surgical-site infection). Relevance: low-medium - shows optical counters as the performance metric for portable cleaners in clinical rooms.

doi: 10.1016/j.infpip.2026.100583

Pigakis cement dust small airway dysfunction CPET 2026 Int Arch Occup Environ Health

Early respiratory effects of long-term cement dust exposure: association with exercise capacity in asymptomatic workers

500 non-smoking, non-atopic cement workers aged >=35 with >=10 years' exposure underwent spirometry, lung volumes, impulse oscillometry and CPET. A subset had small airway dysfunction despite normal spirometry; they had 19.4% lower VO2 peak (29 vs 36 mL/kg/min) and lower breathing reserve (34% vs 43%). Limitation: no unexposed comparator and no measured dust concentrations, so this shows SAD is associated with exercise limitation, not that cement dust causes either; the SAD prevalence is not given. Relevance: low-moderate - occupational dust surveillance still relies on tenure, which personal PM monitors could replace.

doi: 10.1007/s00420-026-02247-x

Stange SEM-Raman inhalable particles indoor artificial turf infill 2026 Research Square (preprint)

Correlative SEM–Raman Microscopy of Inhalable Particles Sampled during Indoor Football on Artificial Turf with Crumb Rubber and Olive-Stone Infills

Air sampling during matches in two Norwegian indoor halls, one with crumb-rubber infill and one with olive-stone/quartz infill, analysed by correlative SEM, AI particle classification and micro-Raman. Both infills released airborne particles. Olive-stone particles appeared indoors only; tyre rubber appeared indoors and outdoors, implicating traffic as well as infill. PM_{10} was 58.4 $\mu g\,m^{-3}$ in the olive-stone hall vs 6.0 $\mu g\,m^{-3}$ with crumb rubber, and fungal spores were found with the organic infill. Limitation: one hall per infill, so the tenfold PM_{10} difference is confounded by hall ventilation, occupancy and cleaning. Preprint. Relevance: low-medium - substituting a 'sustainable' material changed composition and raised mass.

doi: 10.21203/rs.3.rs-11040166/v1

Suh residential occupant activities airborne microplastic emissions 2026 Build Environ

Microplastics emissions from occupant activities in residential spaces: Experimental quantification and relative importance analysis of particle generation factors

METADATA ONLY. Title and authors only; no abstract retrievable at harvest. The title reports experimental quantification of microplastic particle emissions from occupant activities in residential spaces and a relative-importance analysis of generation factors (activity type, textiles, surfaces presumably). Size range, identification method (FTIR, Raman, pyrolysis GC-MS), emission rates and whether the inhalable fraction was measured cannot be stated. Limitation: unverifiable beyond the title; microplastic counts are method-bound and seldom expressed as PM mass. Relevance: moderate - indoor activity-resolved particle sources are what household LCS see as unexplained spikes; polymer fibres also scatter differently from combustion aerosol.

doi: 10.1016/j.buildenv.2026.115399

Wang electret HVAC filter charge neutralisation wildfire smoke 2026 Environ Sci Technol

Charge-Driven Degradation of Residential Electret HVAC Filters under Wildfire Smoke Exposure.

Laboratory flaming and smouldering wood smoke, field sampling at prescribed burns and controlled soot challenges on residential electret HVAC filters. Filtration efficiency collapsed at particle loadings far below those that raise pressure drop; the driver was soot-rich particles neutralising electret charge irreversibly, not mechanical clogging or organic-aerosol deposition. Built into a dynamic indoor model driven by EPA monitor data for a regional wildfire event, effective filter life fell from months to days at moderate outdoor $PM_{2.5}$, with protection varying by building and ventilation. No efficiency numbers are given in the abstract. Limitation: few filter products and smoke types; charge loss in the field depends on soot fraction, which varies by fire stage. Relevance: high - indoor/outdoor LCS pairs can detect the efficiency collapse that pressure-based replacement rules miss.

doi: 10.1021/acs.est.6c03046

Burden, policy & mitigation 14 records

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (14 OF 14 RECORDS IN THIS SUBTOPIC)

Beveridge HeatRisk AQI cardiovascular action algorithm 2026 Am J Prev Cardiol

HeatRisk and air quality risk recommendations: A narrative review and action algorithm for the care of people at risk for and living with cardiovascular disease

PubMed-based narrative review (2000-2025) prioritising systematic reviews, mapped onto CDC, EPA and NWS guidance to build a clinical counselling algorithm for heat and poor air quality in people with or at risk of CVD. It summarises excess risk for heart failure, ischaemic heart disease, stroke and MI, and high-risk groups (pregnancy, older age, certain medications, occupational exposure), and proposes action at a risk score >=3. Limitation: non-systematic and the scoring algorithm is not validated against outcomes. Relevance: moderate - AQI-triggered clinical advice is only as good as the local $PM_{2.5}$ behind it; sensor-informed AQI (AirNow Fire and Smoke map) is the practical feed.

doi: 10.1016/j.ajpc.2026.101704

Brunori Rome urban vineyard grapevine leaf PM interception 2026 Plants (Basel)

Plant-Atmosphere Interactions in Urban Vineyards of Rome: A Digital Phenotyping Approach for the Functional Selection of Grapevine Cultivars

Urban vineyard of 170 cultivars at Rome's Botanical Garden, 2025 season: 54 cultivars screened by leaf micromorphology, five assessed in depth (anatomy, chlorophyll, reflectance indices, SEM-counted PM). After ~20 days without heavy rain, leaf-surface PM ranged from 52,024 (Barbera) to 77,462 particles/mm2 (Maturano), comparable to some urban tree species; adaxial deposition was over twice abaxial. Limitation: particle counts, not mass or size-resolved deposition flux; five cultivars, one site, one accumulation window, and no ambient PM measurement to convert counts into a removal rate. Relevance: low - green-infrastructure mitigation claims need co-located ambient PM, which a low-cost sensor at canopy height would supply cheaply.

doi: 10.3390/plants15193014

Cantu air quality across the lifespan nursing response review 2026 Nurs Clin North Am

Breathing Unequal Air: Global and US Air Quality Across the Lifespan and the Nursing Response.

Narrative review for nurses of global and US air quality trends and health effects across the lifespan, citing ~8 million deaths/year attributable to air pollution, with $PM_{2.5}$, O_3 and NO_2 as the principal pollutants. Covers cardiopulmonary, neurocognitive and perinatal outcomes, nursing practice and policy, and argues for integrating air-quality awareness into nursing curricula. No new data. Limitation: non-systematic; no synthesis of effect sizes. Relevance: low - possible audience for patient-facing use of public sensor maps.

doi: 10.1016/j.cnur.2026.06.007

He ozone-dominant air pollution risk ageing inequality China counties 2026 Ecotoxicol Environ Saf

A nationwide effect-based assessment of ozone-dominant air pollution risk patterns and ageing-related inequality in China

Interpretable machine learning was fit to cancer data from 813 Chinese counties (2018-2019) to get non-linear long-term exposure-response shapes for cold-season $PM_{2.5}$ and warm-season O_3. Model-based thresholds then anchored exceedance and excess-hazard indices projected over 2017-2023. $PM_{2.5}$ pressure fell while O_3 pressure rose, with hotspots in Sichuan-Chongqing and the Yangtze River Delta that coincide with ageing. Limitation: two years of county cancer data cannot identify a dose-response threshold. ML-derived 'thresholds' in an ecological panel are model artefacts confounded by region, and the abstract gives no effect sizes or CIs. Relevance: low - county-level modelled exposure.

doi: 10.1016/j.ecoenv.2026.120945

Idowu medical waste incinerator airborne heavy metals Monte Carlo Nigeria 2026 Research Square (preprint)

Monte Carlo-Based Probabilistic Health Risk Assessment of Airborne Heavy Metals from a Medical Waste Incinerator in Southwestern Nigeria

Particulate matter collected on passive glass slides around a medical-waste incinerator for two months; ten metals analysed, converted to airborne concentrations, and propagated through 10,000 Monte Carlo draws. Median HQ ~400, 95th percentile 850; exceedance probability near 100%; ILCR above benchmarks. Limitation: passive deposition slides measure settled mass, not airborne concentration - the conversion to C_air requires an assumed deposition velocity that dominates the result, and an HQ of 400 almost certainly reflects that conversion rather than real inhalation dose. Preprint. Relevance: moderate as a cautionary example - fence-line LCS or low-volume samplers would have given a defensible air concentration.

doi: 10.21203/rs.3.rs-11381994/v1

Jiang PLA melt-blown nonwoven filtration durability humidity airflow 2026 Langmuir

Service Durability of PLA Melt-Blown Nonwovens: Roles of Humidity and Airflow Velocity in Degradation and Filtration Performance

Full-factorial service-condition study (humidity, airflow velocity, PM loading, natural microbes) on PLA melt-blown filter media. Degradation stays early-stage, first-order, k up to 3.88e-2 /h, surface-erosion dominated. Humidity drives degradation (90% of short-term water-contact-angle variance; 77% of yield-strength and 91% of pressure-drop variance long-term); airflow velocity governs dynamic filtration via particle-media contact. Limitation: the abstract gives no filtration efficiency numbers by particle size, so it is unclear whether MPPS performance falls within a realistic mask or HVAC service life. Relevance: low - mitigation materials; humid-climate filter life matters for purifier intervention trials.

doi: 10.1021/acs.langmuir.6c03984

Liang emergency haze controls causal ML synthetic control Shijiazhuang 2026 Environ Sci Technol

Evaluating Emergency Emission Controls During an Unprecedented Regional Haze Event Using Causal Machine Learning

Three-step causal evaluation (meteorological normalisation, augmented synthetic control, ML prediction) of Shijiazhuang's 45-day 'Sharp Sword' campaign in 2016. Local emission strength fell 4.3% ($PM_{2.5}$), 4.8% (PM_{10}), 15.8% (SO_2), 5.8% (NO_2), 15.3% (CO); concentrations fell 42.1 $\mu g\,m^{-3}$ (14.8%) for $PM_{2.5}$ and 47.4 $\mu g\,m^{-3}$ (12.4%) for PM_{10}. Independent PMF showed vehicle, steelmaking and coal contributions 4.1%, 11.0% and 21.7% lower than in 2017. $PM_{2.5}$ still averaged 243.2 $\mu g\,m^{-3}$ during the campaign. Limitation: donor-city synthetic controls share the regional haze and possibly spillover controls, biasing the effect toward zero; no uncertainty intervals in the abstract. Relevance: medium - a transferable counterfactual framework for evaluating interventions from network data.

doi: 10.1021/acs.est.6c12991

Polwiang Bangkok Chiang Mai burning control DLNM GAM $PM_{2.5}$ regimes 2026 Atmos Pollut Res

Differential sensitivity of urban and biomass-dominated $PM_{2.5}$ regimes to burning control: A distributed-lag generalized additive model analysis of Bangkok and Chiang Mai (2021-2024)

METADATA ONLY. Title and author only; no abstract retrievable at harvest. The title states a distributed-lag GAM comparing the sensitivity of $PM_{2.5}$ to burning-control measures in an urban regime (Bangkok) and a biomass-burning-dominated regime (Chiang Mai), 2021-2024. Effect sizes, lag structure, the burning-control variable (fire counts, policy dates) and confounder set cannot be stated. Limitation: observational policy attribution in a GAM is sensitive to meteorological adjustment and transboundary smoke from Myanmar and Laos; single author. Relevance: moderate - Chiang Mai is a dense PMS-sensor setting (DustBoy network), and regime-specific response is where LCS spatial coverage tests policy effects.

doi: 10.1016/j.apr.2026.103236

Prieto Perez Huszar future central European air quality emission changes 2026 Atmos Environ

The role that changes in emissions play in future central European air-quality

METADATA ONLY. Title and authors only (Prieto Perez, Huszar, Karlicky); no abstract retrievable at harvest. The group runs regional climate-chemistry (RegCM/WRF-CAMx) scenarios; the title separates the emission-change contribution to future central European air quality from climate change. Pollutants, scenarios and horizons cannot be stated. Limitation: unverifiable beyond the title; scenario attribution depends on the emission pathway chosen. Relevance: low - policy context for the PM levels European LCS networks will measure.

doi: 10.1016/j.atmosenv.2026.122404

Scott equine air quality index Delphi management 2026 J Am Vet Med Assoc

Experts recommend management strategies for horses during poor air quality

A three-round modified Delphi with 15 equine veterinarians, 11 of whom completed all rounds. The panel adapted the EPA AQI public messaging to equine health risk and set management, training and competition recommendations for each AQI category. Experts agreed that healthy horses as well as compromised ones can have short- and long-term effects. Limitation: 11 completers, and the recommendations rest on translational extrapolation from human data, as the authors acknowledge. There are no equine exposure-response data. Relevance: medium - venue-level LCS readings at stables and competition grounds are what would operationalise AQI-tiered decisions.

doi: 10.2460/javma.26.05.0436

Warminski SRC pellets boiler ESP PM CO emissions 2026 Molecules

Short-Rotation Coppice Pellets as Biofuels in a Modern Boiler with an Electrostatic Precipitator: Particulate Matter and Carbon Monoxide Emissions

Short-rotation coppice pellets (willow, poplar, black locust) versus pine and birch in a pellet boiler with an electrostatic precipitator. Species significantly affected PM and CO; the ESP cut mean PM by 52.5% (53.81 to 25.54 mg/m3) without affecting CO. Highest PM: willow without ESP (90.96 mg/m3); lowest: pine with ESP (17.51). Poplar gave low PM but the highest CO; black locust the best balance (combustion efficiency 93.20%). Limitation: one boiler, total PM only (no size distribution or PM_1 fraction, where residential-combustion mass sits), and steady-state operation that misses start-up peaks. Relevance: low - source-side mitigation; a 52% ESP removal at total-mass level likely overstates sub-micron removal.

doi: 10.3390/molecules31193443

Zhang Jiaxing inventory co-control screening $PM_{2.5}$ GHG 2026 Atmosphere

Inventory-Based Co-Control Screening of Air Pollutants and Greenhouse Gases in a Typical Yangtze River Delta City

Harmonised 2024 integrated inventory plus enterprise statistics for Jiaxing; 54 non-zero source categories scored on an air-pollutant H-score (mean of $PM_{2.5}$, NOx, SO_2, VOCs, NH_3 shares) against a GWP100 CO_2-eq G-score, screened at the 70th percentile. Coupled co-control candidates: CHP, thermal power, motor vehicles; decoupled single-pollutant priorities: industrial solvents (VOCs) and paved-road dust (PM). A 75th-percentile benchmark implies 7,941 t VOC reduction among reporting enterprises. Limitation: equal-weight averaging of pollutant shares ignores toxicity and secondary PM formation potential; no ambient or health validation. Relevance: low - road-dust PM priority is the coarse fraction optical LCS undercount.

doi: 10.3390/atmos17101023

Zhang early-onset female lung cancer burden GBD 2021 2026 Asia Pac J Clin Oncol

Global Burden and Trends of Early-Onset Tracheal, Bronchus, and Lung Cancer Among Women From 1990 to 2021.

GBD 2021 secondary analysis of early-onset (under 50) female TBL cancer, 1990-2021. Global rates declined (all EAPCs <0), but low-middle SDI regions rose (DALY EAPC 0.66) and South Asia fastest (DALY EAPC 1.42). Ambient PM was the fastest-growing attributable risk (attributable DALY rate EAPC 1.22, ASDR 1.27). Limitation: GBD attributable burden is model output that inherits the exposure surface and IER curves; in young women with few cases, estimates in low-SDI regions rest largely on modelled covariates rather than registries. Relevance: low - burden, not measurement.

doi: 10.1111/ajco.70212

Zhang respirable magnetite emissions coal-fired power China 2026 Environ Sci Technol

Emissions of Respirable Magnetite from the Coal-Fired Power Sector in China: Spatial Inequalities and Mitigation Pathways

Multifactor emission model for respirable magnetite from Chinese coal-fired power: 2,828.7 (2,307.1-3,371.4) t/y, spatially concentrated by tiered emission standards. Guizhou and Sichuan supply 4% of coal power but tightening their standards alone would cut >31% of national magnetite emissions. Limitation: emission factors for magnetite specifically (as opposed to total PM or Fe) rest on few stack measurements, and the uncertainty interval looks narrow for a species-level inventory; health relevance is asserted from brain-magnetite literature rather than exposure-response data. Relevance: moderate - magnetite is invisible to optical LCS mass; magnetic or filter-based speciation would be needed to verify hotspots.

doi: 10.1021/acs.est.6c08602

Other clinical endpoints — 5 records

Carried from the daily issues — full summaries (5 of 5 records in this subtopic)

Bottini antibiotic resistance Ecoli Salmonella $PM_{2.5}$ ecological 2026 One Health

Environmental, socioeconomic, and livestock-related factors associated with antibiotic resistance in Escherichia coli, Shigella spp., and Salmonella enterica

397,103 isolates from human, animal and environmental sources (NCBI Pathogen Detection, 2003-2018) in mixed-effects negative binomial models. For E. coli/Shigella, $PM_{2.5}$ was associated with beta-lactam resistance (PRR 1.39, 1.11-1.74), alongside low income and population density; for S. enterica, $PM_{2.5}$ and tetracycline resistance PRR 1.65 (0.98-2.76), with warm climate and poultry density. Limitation: national $PM_{2.5}$ is a marker of industrialisation, sanitation and antibiotic use, and sequencing effort varies by country; the association is not interpretable as a PM effect. Relevance: low - included because $PM_{2.5}$ is reported with an estimate; the airborne-resistome hypothesis would need measured bioaerosol, not country means.

doi: 10.1016/j.onehlt.2026.101600

Jiang air pollution meteorology hand foot mouth disease Shenzhen DLNM 2026 Front Public Health

Ambient air pollution and meteorological factors associated with hand, foot, and mouth disease risk in Shenzhen, China: a time-series study

Shenzhen daily HFMD counts 2013-2022 with six pollutants and meteorology in DLNMs, including pre/post EV71 vaccine and COVID-19 periods. $PM_{2.5}$ and PM_{10} showed risk peaks at low concentrations and 'protective' effects at extreme highs; SO_2 and CO raised cumulative risk at extremes; temperature showed a dual-peak pattern and high humidity raised risk. No effect sizes with CIs in the abstract. Limitation: HFMD is a contact-transmitted enteroviral disease with strong seasonality driven by school terms; non-monotone, sign-flipping PM curves are the signature of residual seasonal confounding and DLNM over-flexibility, not biology. Relevance: low.

doi: 10.3389/fpubh.2026.1920424

Lin $PM_{2.5}$ frailty trajectories CHARLS 2026 J Occup Environ Med

The association of particulate matter 2.5 exposure with frailty trajectories in middle-aged and older Chinese adults and the potential modification by lifestyle and socioeconomic status

10,659 CHARLS participants; deficit-accumulation frailty index, three GBTM trajectories (stable low 64%, increasing to frailty 29%, continuously worsening 7%). Per 10 $\mu g\,m^{-3}$ $PM_{2.5}$, odds of the two adverse trajectories rose 15.2% and 12.3% (ordered logistic; no CIs in the abstract). Higher SES attenuated the association (RERI -0.067); unhealthy sleep duration amplified it (RERI 0.163). Limitation: GBTM classes are model artefacts with posterior misclassification carried into a second-stage regression without uncertainty propagation; CHARLS addresses are coarse (city/county), so exposure contrast is mostly between regions with different healthcare and diet. Relevance: low - regional satellite exposure, no sensing implication.

doi: 10.1097/jom.0000000000003913

Lin air pollution cervical cancer recurrence mortality Fudan cohort 2026 Environ Health (ACS)

Long-Term Exposure to Ambient Air Pollution and Cervical Cancer Outcomes in Chinese Women: A Prospective Study on Recurrence and Mortality Risks

3,758 women with primary cervical cancer (2019-2022), median follow-up ~3 y; 217 recurrences, 113 deaths. Per IQR $PM_{2.5}$: recurrence SHR 1.93 (1.40-2.66, Fine-Gray), death HR 2.55 (1.78-3.67); PM_{10} SHR 1.65 (1.13-2.42) and HR 2.21 (1.38-3.54); NO_2 and SO_2 also positive, O_3 null. Limitation: hazard ratios near 2-2.5 per IQR from 113 deaths are implausibly large for a residential long-term contrast and point to confounding by referral geography - patients from outside Shanghai (higher PM) likely differ in stage, HPV management and follow-up intensity; single-centre, short follow-up, no adjustment for distance to the hospital reported in the abstract. Relevance: low - model-based residential exposure, no sensing implication.

doi: 10.1021/envhealth.6c00011

Wen landscape fire $PM_{2.5}$ renal hospitalisations six-country time-series 2026 Environ Sci Technol

Landscape Fire-Sourced $PM_{2.5}$ and Cause-Specific Renal Hospitalizations: A Multicountry Two-Stage Time-Series Analysis of 12.5 Million Admissions

12.5 million renal admissions across 1020 locations in six countries, 2000-2019; two-stage time-series with fire-sourced $PM_{2.5}$ separated from the non-fire remainder. Per 10 $\mu g\,m^{-3}$ cumulative lag 0-5, RR 1.011 (1.007-1.015) for all renal admissions, strongest for acute kidney injury; associations larger for fire-sourced than non-fire $PM_{2.5}$, larger in warmer areas and graded by GDP per capita. Attributable burden 1.38 admissions per 100,000 per year. Limitation: the fire/non-fire split comes from a chemical-transport-plus-ML partition, not measured tracers, so 'fire PM is more toxic' partly inherits the partition's error structure; heat co-occurs with smoke and dehydration drives AKI, and residual confounding by temperature is the obvious threat. Relevance: moderate - smoke-episode exposure is where LCS density most improves over sparse FRM/FEM coverage.

doi: 10.1021/acs.est.6c04677

Pooled corpus register

First author	Focus	Journal	Design	Metric	Region
Neuro / mental health					
Baird woodsmoke CD4 T cell brain influx saturated fat diet mice	CD4+ T Cell Activation and Peripheral Immune Cell Influx into the Brain Following Wildfire Smoke Exposure is Modulated by a Saturated Fat Diet	*Neurotoxicology*	Animal exposure study	Woodsmoke at 0.5 mg/m3, 4 h every other day for 14 days	Not applicable (laboratory)
Kalkbrenner traffic air pollution ADHD Denmark critical windows	Traffic-related air pollution links with attention deficit hyperactivity disorder in Denmark: Developmental periods of susceptibility and sex-specific associations	*Environ Res*	Case-cohort study	Daily address-level $PM_{2.5}$ and elemental carbon (DEHM/UBM/AirGIS), pre-conception to age 12	Denmark (national registers)

First author	Focus	Journal	Design	Metric	Region
Wang multi-omics Mendelian randomization $PM_{2.5}$ depression	Multi-omics Mendelian Randomization Reveals the Causal Association and Mediating Mechanisms of $PM_{2.5}$ with Depression: A Systematic Study Integrating Transcriptomics, Metabolomics, and Radiomics	*Research Square (preprint)*	Two-sample Mendelian randomisation	Genetically predicted $PM_{2.5}$ exposure (GWAS instruments)	Global / multi-region (GWAS summary data)
Zhang CLHLS $PM_{2.5}$ components cognitive disorders lifestyle mediation	Association between long-term ambient air pollution exposure and cognitive disorders among older adults in China: the mediating role of lifestyle	*J Glob Health*	Prospective cohort	Long-term $PM_{2.5}$, sulphate, nitrate, ammonium, O3 (per 10 $\mu g/m^3$)	China (CLHLS, multi-province)
Cardiovascular & metabolic					
Brugge HEPA sham crossover cytokines visceral adiposity	Pro-inflammatory cytokines moderated by visceral adiposity from a pragmatic randomized crossover trial of portable HEPA air purifiers	*Am J Prev Cardiol*	Randomised crossover trial	Indoor PM reduction by portable HEPA vs sham filtration (30-day periods)	Not stated in abstract
Du Air Health Index acute myocardial infarction 325 cities	Establishing an Air Health Index for Predicting the Risk of Acute Myocardial Infarction Onset Associated with Air Pollution and Ambient Temperature	*Environ Sci Technol*	Health impact model	Daily $PM_{2.5}$ (with NO2, O3, temperature) from high-resolution surfaces	China (325 cities)
Hahad exposome heart failure older adults review	Exposomic Determinants in Heart Failure Among Older Adults - Mechanistic Links, Clinical Evidence, and Future Perspectives	*Curr Heart Fail Rep*	Narrative review	Air pollution (PM) within a multi-exposure exposome (noise, heat, metals, light at night, greenness)	Global / multi-region
Huang $PM_{2.5}$ coronary artery dilatation Kawasaki disease case-crossover	Association between PM(2.5) and coronary artery dilatation with Kawasaki disease in China	*Eur J Pediatr*	Case-crossover	Short-term $PM_{2.5}$ (distributed lag 0-7 d)	China (single paediatric centre)
Lee CRCS-K post-discharge SO2 stroke recurrence PM null	Post-Discharge Ambient Sulfur Dioxide Exposure and Stroke Recurrence After Acute Ischemic Stroke	*J Stroke*	Prospective cohort (registry linkage)	3-month post-discharge district PM_{10}, $PM_{2.5}$, NO2, SO2, CO, O3 (IDW)	South Korea (CRCS-K-NIH registry)
Rao $PM_{2.5}$ components cardiorenal multistate trajectory	Associations of $PM_{2.5}$ components with dynamic and accelerated progression of different kidney function states and subsequent cardiovascular events	*Int J Cardiol*	Prospective cohort	2-year $PM_{2.5}$ and components: sulfate, nitrate, ammonium, organic matter, black carbon	Not stated in abstract (cohort of 307,639)
Sun long-term air pollution lipid profiles Shanxi repeated measures	Long-term exposure to ambient air pollution and lipid profiles: a longitudinal study in a heavily industrialized region of northern China	*Front Public Health*	Prospective cohort	3-y mean $PM_{2.5}$, PM_{10} (ChinaHighAirPollutants, geocoded residence)	China (Shanxi Province)
Wan $PM_{2.5}$ mortality urban rural elderly Beijing cohort	Association between fine particulate matter exposure and all-cause and cardiovascular mortality in urban and rural elderly population in Beijing	*Zhonghua Liu Xing Bing Xue Za Zhi*	Prospective cohort	Satellite-gridded annual $PM_{2.5}$ at residence (2010-2014 mean 79.8 $\mu g/m^3$)	China (Beijing urban and rural)
Wongcharoen short-term $PM_{2.5}$ MACE high-risk population nationwide case-crossover	Associations between short-term PM(2.5) exposure and major cardiovascular events in high-risk population: a nationwide case-crossover study	*BMJ Open*	Metadata only (no abstract)	Short-term $PM_{2.5}$	Thailand (inferred from author group; not stated in record)
Yadav PM pollution-accelerated cardiovascular aging PACA framework	Particulate Matter Exposure and Accelerated Cardiovascular Aging: Mechanistic Evidence, Biomarker Integration, and the PACA Framework	*Cardiovasc Toxicol*	Mechanistic review	$PM_{2.5}$ and ultrafine particles (literature)	Global / multi-region (review)
Zhang wildfire smoke waves long-term $PM_{2.5}$ cardiovascular hospitalisation Medicare	Impact of wildfire smoke waves on the association between long-term exposure to fine particulate matter and cardiovascular hospitalizations among older adults in the United States	*Environ Res Health*	Nationwide cohort	Two-year mean $PM_{2.5}$ (1 km ML model) and smoke $PM_{2.5}$ (10 km); smoke wave = >=2 d with smoke $PM_{2.5}$ >10 $\mu g/m^3$	USA (contiguous)
Zhu source-specific $PM_{2.5}$ atrial fibrillation Medicare cohort	Source-Specific $PM_{2.5}$ Exposure and Atrial Fibrillation Risk in U.S. Older Adults: A Nationwide Cohort Study	*Environ Health (ACS)*	Retrospective cohort	Source-apportioned $PM_{2.5}$ (Multilinear Engine on CSN/IMPROVE speciation; IDW to ZIP code, 100 km)	USA (nationwide Medicare)

First author	Focus	Journal	Design	Metric	Region
Reproductive & developmental					
Atkin air quality excess heat pregnancy outcomes review	Impact of Poor Air Quality and Excess Heat on Pregnancy Outcomes.	*Nurs Clin North Am*	Narrative review	Ambient air pollution and heat (reviewed)	Not stated in record
Falcon-Gracia BiSC placental sncRNA air pollution maternal smoking	Impact of air pollution and tobacco smoke on the placental small non-coding RNA landscape	*medRxiv (preprint)*	Cross-sectional (birth cohort)	Prenatal $PM_{2.5}$, black carbon, NO2 (hybrid dispersion-LUR)	Spain (Barcelona, BiSC)
Ge preconception folic acid air pollution birthweight	Preconception Folic Acid Supplementation Associated with Attenuated Effects of Air Pollution on Birthweight Outcomes	*Environ Health (ACS)*	Metadata only (no abstract)	Ambient air pollution (metrics not stated in record)	China (inferred from author group; not stated in record)
Kalam HAPIN LPG intervention child growth stunting ages 2-5	Effects of a 500-day LPG cookstove and fuel intervention delivered during gestation and infancy on height-for-age (HAZ) and stunting at ages 2, 3, 4 and 5 years	*medRxiv (preprint)*	Trial / intervention	Personal $PM_{2.5}$, CO and black carbon (gestation, infancy, post-trial)	Guatemala, India and Rwanda (HAPIN 2.0 follow-up)
Liu $PM_{2.5}$ components small for gestational age IVF Guangzhou	Maternal exposure to PM(2.5) and its components and risk of small for gestational age in women undergoing in vitro fertilization: a propensity score-matched case-control study in Guangzhou, China.	*BMC Public Health*	Case-control study	Trimester-averaged $PM_{2.5}$ and SO4, NO3, NH4, OM, BC (gridded estimates at residence)	China (Guangzhou)
Moradnia Isfahan pregnant urinary metals indoor dust AQI	Spatial variability of heavy metals in pregnant women: linking urinary biomarkers, indoor dust contamination, and air quality in Isfahan, Iran	*Environ Geochem Health*	Cross-sectional environmental survey	Zone-level Air Quality Index; indoor settled dust metals (not airborne PM)	Iran (Isfahan)
Respiratory & allergic					
Argersinger air pollution chronic rhinosinusitis review	The link between air pollution and chronic rhinosinusitis	*Curr Opin Otolaryngol Head Neck Surg*	Narrative review	$PM_{2.5}$ (traffic, industrial, wildfire, pollen sources)	Global / multi-region (review)
Gonehgar sand dust storms asthma Asia systematic review	Geographic Patterns of Asthma Morbidity: A Systematic Review of Sand and Dust Storm (SDS) Impacts in Asia	*J Asthma*	Systematic review	Dust-event PM_{10} and $PM_{2.5}$ (study-specific definitions)	Asia (Japan, Korea, Middle East)
Guerra-Ruiz cytokines volcanic emissions Cumbre Vieja ASHES	Systemic cytokine signatures associated with volcanic emisions exposure and respiratory disease after the cumbre vieja eruption-ashes project.	*Ecotoxicol Environ Saf*	Observational - cross-sectional	Volcanic gas and particulate exposure (hours of exposure, group membership)	Spain (La Palma)
Harb idiopathic subglottic stenosis geographic clustering ecological	Geographic Clustering of Idiopathic Subglottic Stenosis in North America: An Ecological Cohort Study	*Laryngoscope Investig Otolaryngol*	Ecological	State/province $PM_{2.5}$ alongside temperature, humidity and elevation	North America (USA and Canada)
Jung peaker power plant emissions ED urgent care visits Massachusetts	Individual-level health effects of living near a gas-fired "Peaker" power plant	*J Clim Change Health*	Case-crossover	HyADS trajectory-derived peaker-plant exposure index (unitless); background $PM_{2.5}$ adjusted	USA (Massachusetts)
Kim interstitial lung disease changing climate Personal View	Interstitial lung diseases in a changing climate: emerging risks and opportunities to improve care	*Lancet Respir Med*	Narrative review	Air pollution and wildfire smoke among climate exposures (literature)	Global / multi-region (review)
Li size-specific PM lung function China Pulmonary Health Study	Associations between Size-Specific Particulate Matter and Lung Function: Results from the China Pulmonary Health Study	*Environ Health (ACS)*	Observational - cross-sectional	Long-term PM1, PM1-2.5, $PM_{2.5}$, $PM_{2.5}$-10, PM_{10} (per IQR)	China (national, CPH Study)
Meng traffic noise air pollution lung function randomised crossover	Independent and interactive effects of traffic-related noise and air pollution on lung function: A randomized crossover study in healthy young adults.	*Environ Res*	Randomised crossover trial	Roadside PM1, $PM_{2.5}$, PM_{10}, O3, NO2 and noise during evening rush hour	China (urban roadside)
Morales El Paso port-of-entry $PM_{2.5}$ immigration COPD census tracts	Breathing at the Border: $PM_{2.5}$ Exposure, Immigration, and COPD Across Census Tracts Close to Ports-of-Entry	*J Immigr Minor Health*	Ecological / registry analysis	Census-tract $PM_{2.5}$ (NASA satellite-derived)	USA (El Paso, Texas, US-Mexico border)
Tullis cystic fibrosis CRS olfaction pollutants	Environmental Exposures, Chronic Sinusitis, and Olfaction in People With Cystic Fibrosis	*Int Forum Allergy Rhinol*	Cross-sectional (cohort baseline)	3-y mean $PM_{2.5}$, PM_{10} (and gases) from nearest EPA monitor to residential ZIP	United States (multicentre)

First author	Focus	Journal	Design	Metric	Region
Wang Guangdong $PM_{2.5}$ oxidative potential DTT childhood allergy	Associations of $PM_{2.5}$ and Its Oxidative Potential with Childhood Allergy-Related Symptoms: A Cross-Sectional Study in Guangdong Province, China	*Environ Sci Technol*	School-based cross-sectional	$PM_{2.5}$ mass, oxidative potential (DTTv, DTTm), metals, carbonaceous, ions	China (Guangdong, six cities)
Mechanistic toxicology					
Amato mobile air-liquid interface toxicity Barcelona harbor urban	Assessment of air pollution toxicity in urban environments in Barcelona, Spain, using a mobile air-liquid interface system	*Environ Sci Pollut Res Int*	Exposure assessment + in vitro	Co-measured particle and gas concentrations; whole-air ALI exposure	Spain (Barcelona harbour and urban background)
Jiang carbon nanoparticle NSCLC progression ARNT2 ARL14	Carbon nanoparticle exposure promotes the progression of non-small cell lung cancer via the ARNT2/ARL14 axis: Findings from in vivo and in vitro models	*Ecotoxicol Environ Saf*	Animal + in vitro	Carbon nanoparticles (intranasal 15-200 ug/kg; in vitro 2 ug/mL)	China (laboratory)
Kou Xing Su San Sang Xing Tang $PM_{2.5}$ lung injury rats	Xing Su San and Sang Xing Tang attenuate $PM_{2.5}$-induced lung injury and inflammatory marker expression in rats	*Pak J Pharm Sci*	Animal exposure study	$PM_{2.5}$ inhalation 500 $\mu g/m^3$, 3 h/day, 28 days, under warm-dry or cool-dry conditions	China (laboratory)
Liu $PM_{2.5}$ IL-17 microglia neuroinflammation	IL-17 Signaling Pathway: A Potential Signaling Pathway Mediating Neuroinflammation Induced by $PM_{2.5}$	*Environ Health (ACS)*	Animal + in vitro	Collected $PM_{2.5}$ suspension (dose not stated in abstract)	Laboratory
Ruan gut microbiota dose-time trajectories $PM_{2.5}$ PFOS mouse	Time-resolved profiling reveals nonlinear dose-time trajectories of gut microbiota in $PM_{2.5}$ and PFOS exposure models	*Environ Pollut*	Animal exposure study	$PM_{2.5}$ inhalation at multiple dose levels (separate PFOS drinking-water model)	Not stated
Sun respirable mining dust natural killer cell dysfunction	Respirable Particulate Matter Induces Natural Killer Cell Dysfunction via Oxidative Stress-Mediated Lysosomal and Mitochondrial Injury: Implications for Mineral-Rich Mining Dust Exposure.	*Environ Sci Technol*	In vitro	Compositionally distinct respirable PM (silica/silicate, iron-bearing, carbonaceous mine dusts)	Not stated in record
Tang CPS-II plasma proteome air pollution lung cancer ever-smokers	Analysis of the Potential Mediating Role of Plasma Proteins in the Association Between Air Pollution Exposure and Lung Cancer Risk Among Ever-Smokers	*Environ Sci Technol*	Nested case-control	Six ambient pollutants and their mixture (pre-diagnosis)	USA (CPS-II Nutrition Cohort)
Wang seasonal PM size constituents oxylipin panel Beijing	Winter and Summer Differences in Oxylipin Responses to Ambient Particles: Roles of Sizes and Constituents	*Environ Sci Technol*	Panel study	27 $PM_{2.5}$ metrics: mass, UFP and accumulation-mode number, OC, chloride and other constituents	China (Beijing)
Zendi brown carbon wood pyrolysis tracers cytotoxicity A549	Fine Brown Carbon Aerosol Emitted by Wood Pyrolysis - New Harmful Tracers and Cytotoxicity in Human Lung Cells	*Chem Res Toxicol*	In vitro (cell lines)	Fine brown carbon from 350 C wood pyrolysis; water, methanol and acetone extracts	Not applicable (laboratory)
de Luna wildfire PM toxicity systematic review	Systematic review on the toxicity of wildfire emitted particulate matter: State of science, knowledge gaps and way forward	*J Hazard Mater*	Systematic review (PRISMA)	Wildfire-emitted PM (field-collected or laboratory surrogate)	Global / multi-region
Occupational & indoor					
Alquwayi respirable crystalline silica occupational exposure Saudi Arabia review	Potential risk of occupational exposure to respirable crystalline silica in Saudi Arabia: a comprehensive review of the current status and future challenges	*Front Public Health*	Narrative review	Respirable crystalline silica (respirable dust fraction)	Saudi Arabia
Arif Agra leather footwear facilities indoor $PM_{2.5}$ CO2 formaldehyde	Indoor Air Quality and Thermal Environment Across Three Leather Footwear Facilities in Agra India	*Research Square (preprint)*	Instrument evaluation	Indoor $PM_{2.5}$/PM_{10}, I/O $PM_{2.5}$ ratio, CO2, formaldehyde, TVOC	India (Agra)
Du classroom CO2 proxy PM VOC ventilation types	CO2 as a proxy for PM and VOCs in classrooms across ventilation types and activities	*Indoor Environments*	Metadata only (no abstract)	Classroom PM (fractions not stated) vs CO2 and VOCs	Not stated in record
Kim pig building air cleaner TSP $PM_{2.5}$ productivity	Association between indoor air pollutants and pig productivity index by on-site application of air cleaner	*J Anim Sci Technol*	Intervention field trial (non-randomised)	TSP, $PM_{2.5}$, airborne fungi in pig housing	South Korea (commercial pig farm)

First author	Focus	Journal	Design	Metric	Region
Kuye ultrafine particle respiratory deposition dose wood stoves UK homes	Respiratory Deposition Dose of Ultrafine Particles from Improved Wood Stoves and Open Fireplaces	*Atmosphere*	Field measurement	Indoor particle number concentration (1-min) with ICRP deposition model	United Kingdom (four homes)
Matuskova wood stove phase-resolved particle-bound PAHs	Characterisation of particle-bound PAHs emitted from a residential wood-burning stove and their dependence on combustion phases	*Environ Res*	Chamber emission characterisation	Cascade-impactor size-resolved PM and particle-bound PAHs (GC-MS/MS)	Central Europe (laboratory stove; beech, black locust, spruce)
Mohammed mobile HEPA-UV air decontamination operating theatre	Evaluation of a mobile surgical air decontamination device in reducing airborne microbial load in the operating theatre of a tertiary care hospital.	*Infect Prev Pract*	Intervention field trial (non-randomised)	Optical particle counts (0.3, 0.5 um and larger) and culturable CFU	India (New Delhi)
Pigakis cement dust small airway dysfunction CPET	Early respiratory effects of long-term cement dust exposure: association with exercise capacity in asymptomatic workers	*Int Arch Occup Environ Health*	Cross-sectional (occupational)	Cement dust (>=10 y occupational tenure; no measured concentrations in abstract)	Not stated in abstract
Stange SEM-Raman inhalable particles indoor artificial turf infill	Correlative SEM–Raman Microscopy of Inhalable Particles Sampled during Indoor Football on Artificial Turf with Crumb Rubber and Olive-Stone Infills	*Research Square (preprint)*	Field study	PM_{10} mass; single-particle SEM + AI classification + micro-Raman of airborne dust	Norway (two indoor sports halls)
Suh residential occupant activities airborne microplastic emissions	Microplastics emissions from occupant activities in residential spaces: Experimental quantification and relative importance analysis of particle generation factors	*Build Environ*	Metadata only (no abstract)	Airborne microplastic particles from occupant activities	Not stated in record
Wang electret HVAC filter charge neutralisation wildfire smoke	Charge-Driven Degradation of Residential Electret HVAC Filters under Wildfire Smoke Exposure.	*Environ Sci Technol*	Materials / filtration	Flaming/smouldering wood smoke, prescribed-burn smoke and soot challenges; EPA monitor $PM_{2.5}$ for model inputs	United States (laboratory, prescribed burns, regional wildfire scenario)
Sensing, forecasting & instrumentation					
Birinci Bosphorus bridge traffic ML proxy $PM_{2.5}$ NO2	Machine-Learning Proxy of Air-Quality Variability Associated with Intercontinental Bridge Traffic over the Bosphorus in Istanbul	*Atmosphere*	Machine-learning model	Hourly $PM_{2.5}$, PM_{10}, coarse fraction, NO2, NO, NOx	Turkiye (Istanbul)
Cai self-supervised stacking particulate nitrate sparse observations	A Self-supervised Stacking Framework for Estimating Particulate Nitrate in Two Chinese Urban Agglomerations under Sparse Ground Observations	*Atmos Environ*	Metadata only (no abstract)	Particulate nitrate (estimated from sparse ground observations)	China (two urban agglomerations)
Cover solvatochromic fluorophores in situ aerosol fluorescence	Evaluation of solvatochromic fluorophores for in situ fluorescence characterization of aerosols	*Aerosol Sci Technol*	Metadata only (no abstract)	Aerosol particles (in situ fluorescence probes)	Not stated in record
Ghaffarpasand Pope vehicle remote sensing real-world emissions high emitters	Emerging Evidence on Real-World Exhaust Emissions and High Emitting Vehicles from Remote Sensing Observations	*Atmos Pollut Res*	Metadata only (no abstract)	On-road remote sensing of exhaust (pollutants not stated in record)	Not stated (review)
Huang foliar particulate UV-Vis-NIR calibration transfer	Nondestructive UV-Vis-NIR assessment of foliar particulate retention in urban vegetation: cross-species transfer and calibration correction	*Environ Monit Assess*	Spectroscopic calibration-transfer study	Leaf-surface particulate mass per area (washing-filtration reference)	China (Nanjing)
Intra four co-deployed electrostatic PM monitors power plant agreement	Multi-point monitoring of PM_{10}, $PM_{2.5}$, and PM1.0 at a combined-cycle power plant: field agreement between four co-deployed electrostatic monitors	*Research Square (preprint)*	Instrument evaluation	PM_{10}, $PM_{2.5}$, PM1.0 (corona charging, impaction, Faraday-cup electrometry)	Thailand (Samut Prakan)
Kheder Foehn generative km-scale global $PM_{2.5}$ forecast	Global probabilistic forecasting of air pollution	*arXiv (preprint)*	Deep-learning forecasting	Global kilometre-scale $PM_{2.5}$ estimates (training and target)	Global
Li runway visual range model brown carbon aerosol chromatic effects	Refined runway visual range measurement model incorporating brown carbon aerosol effects	*Opt Lett*	Theoretical / optical modelling	Brown carbon aerosol spectral extinction	Not stated in record

First author	Focus	Journal	Design	Metric	Region
Pyae Tin Saw portable $PM_{2.5}$ cross-site prediction fixed station	Short-term $PM_{2.5}$ data continuity: Cross-site prediction of portable measurements using a nearby fixed monitoring station	*Environ Pollut*	Metadata only (no abstract)	Portable $PM_{2.5}$ monitor vs nearby fixed-station $PM_{2.5}$	Thailand (inferred from author affiliation; not stated in record)
Ren physics-constrained UQ urban $PM_{2.5}$ prediction driving factors	Integrating Implicit Physics Constraints and Uncertainty Quantification for Urban $PM_{2.5}$ Prediction and Driving Factor Analysis	*Build Environ*	Metadata only (no abstract)	Urban $PM_{2.5}$ (prediction target)	Not stated in record
Risti Jambi PMS5003 PMS7003 vs BAM-1020	Evaluation of Low-Cost $PM_{2.5}$ Sensor Accuracy in Comparison with Government Reference Instrument	*Lingkar: J Environ Eng*	Co-location calibration campaign	Plantower PMS5003 and PMS7003 vs BAM-1020 $PM_{2.5}$	Indonesia (Jambi)
Sharma Singapore aerosol type classification ground remote sensing reanalysis	Aerosol Types Classification in the Tropical Climate of Singapore by Integrating Ground-Based Remote Sensing and Reanalysis Datasets	*Aerosol Air Qual Res*	Metadata only (no abstract)	Columnar aerosol optical properties (AOD, Angstrom exponent presumably)	Singapore
Sim real-time dust monitor bias gravimetric meta-analysis	Systematic bias in real-time dust monitoring: a meta-analysis of measurement discrepancy and its implications for occupational exposure assessment.	*Ann Work Expo Health*	Systematic review + random-effects meta-analysis	Optical real-time dust monitors (DustTrak, SidePak, pDR) vs gravimetric samplers	Global / multi-region (occupational literature 2010-2025)
Srivastav ion particle size distribution NPF Indo-Gangetic Plain	Ion and particle size distribution measurements in the Indo-Gangetic Plain: seasonal insights and challenges in new particle formation event characterization	*Atmos Chem Phys*	Long-term field measurement campaign	Air-ion (0.8-42 nm) and particle (2.5-42 nm) size distributions	India (Sonipat, Haryana)
Wu ProNet physics-AI coupled CTM $PM_{2.5}$ O3 forecasting 338 cities	Fusing physics and AI enhances accuracy and diagnostic capabilities in air quality forecasting	*Natl Sci Rev*	Deep-learning forecasting	$PM_{2.5}$ and O3 forecasts (338 Chinese cities)	China (338 cities)
Exposure assessment & modelling					
Alturki ensemble ML emulation GEOS-CF $PM_{2.5}$ Arabian Peninsula spatial transfer	Interpretable Ensemble Machine Learning for Statistical Emulation of Summer $PM_{2.5}$ Change in GEOS-CF over the Arabian Peninsula: How Far Does Spatially Blocked Skill Transfer?	*Atmosphere*	Machine-learning model	GEOS-CF summer $PM_{2.5}$ relative change 2020-2025 (0.25 deg); TROPOMI, ERA5-Land, CHIRPS, MODIS predictors	Arabian Peninsula
Barnett-Itzhaki hourly compound heat air pollution extremes	Hourly Resolution Reveals Hidden Temporal Windows of Compound Heat-Air Pollution Extremes Masked by Daily Averaging	*Atmos Pollut Res*	Metadata only (no abstract)	Hourly air pollution (pollutant set not stated) co-occurring with heat	Not stated in record
Bloise water-soluble carbon nitrogen PM_{10} Chennai street vertical	Seasonal, Spatial, and Near-Surface Vertical Variability of Water-Soluble Carbon and Nitrogen in PM_{10} in Chennai, India	*Atmosphere*	Field measurement	PM_{10} mass, WSOC, WSTN, WSIN at 3, 16 and 37 m	India (Chennai street sites)
Chen ambient $PM_{2.5}$ representativeness children microenvironments	Evaluating the representativeness of ambient $PM_{2.5}$ for children's exposure and source characteristics across microenvironments during school semesters	*Build Environ*	Metadata only (no abstract)	Ambient vs microenvironmental $PM_{2.5}$ (instrumentation not stated)	Not stated in record
Chen primary brown carbon China 2005-2020 trends radiative effect	Long-term evolution and effects of primary brown carbon aerosol in China	*Atmos Chem Phys*	Chemical transport modelling	Modelled primary brown carbon (internal-mixing optical model), $PM_{2.5}$	China
Cheng LETKF WRF-Chem CALIOP component-resolved aerosol assimilation East Asia	Constraining the vertical profiles of aerosol component and heating rate over east Asia through assimilating CALIOP observations	*Atmos Chem Phys*	Chemical transport modelling	Species-resolved extinction profiles: dust, sea salt, black carbon, water-soluble	East Asia
Dayanandana CMIP6 dust optical depth MENA evaluation	Evaluation of Dust Optical Depth in CMIP6 Simulations over the Middle East-North Africa Region	*Atmos Environ*	Metadata only (no abstract)	Dust optical depth (CMIP6 models vs observations)	Multi-country (Middle East and North Africa)
Ghosh meter-scale heterogeneity new particle formation 3D model	How Microscale Flow and Vapor Transport Shape New Particle Formation and Growth.	*Environ Sci Technol*	Theory / simulation	Simulated nucleation-mode particle number and size distributions (3D-ICAM, 10 m resolution)	Not stated in record

First author	Focus	Journal	Design	Metric	Region
He NYC-METS size-resolved organic aerosol offline AMS	Characteristics of size-resolved organic aerosols using offline aerosol mass spectrometry: NYC-METS	*Aerosol Sci Technol*	Metadata only (no abstract)	Size-resolved organic aerosol (offline AMS)	USA (New York City)
Imran Chinese cities urban air quality determinants interpretable ML 2014-2023	Spatio-temporal variation and determinants of urban air quality in Chinese cities during 2014-2023: an interpretable machine learning analysis	*J Environ Manage*	Metadata only (no abstract)	City-level air quality ($PM_{2.5}$ presumably among pollutants)	China (multi-city)
Kaskaoutis in situ aerosol optical properties eastern Mediterranean	In situ aerosol optical properties in the background eastern Mediterranean: seasonal variability, regional transport, and source attribution	*Atmos Environ*	Metadata only (no abstract)	In situ aerosol scattering and absorption	Not stated in record (eastern Mediterranean)
Kim SpaMixed spatial mixed models microbiome $PM_{2.5}$	Spatial mixed models for assessing environmental exposure effects on the microbiome	*Genome Med*	Statistical methods (simulation + application)	$PM_{2.5}$ (two applied studies)	Not stated (methods paper)
Kirsanov COSMO-ART Saint Petersburg UHI pollution PM_{10} feedback	Assessing Urban Pollution and Heat Island Feedback in a Northern Megacity Using the Fully Coupled COSMO-ART Model	*Atmosphere*	Chemical-transport modelling	Near-surface PM_{10}, NO, CO, O3; aerosol radiative feedback	Russia (Saint Petersburg)
Langford cooking dominant primary PM1 eddy covariance London Beijing Delhi	Traffic and combustion emission controls leave cooking as a dominant local source of primary PM1 in megacities	*npj Clim Atmos Sci*	Eddy covariance flux measurement, method development	Chemically resolved non-refractory PM1 fluxes (aerosol mass spectrometry + eddy covariance)	Global / multi-region (London, Beijing, Delhi)
Li atmospheric cadmium concentration deposition China modelling	Modeling the spatial distributions and seasonal variations of atmospheric cadmium concentration and deposition in China	*Atmos Environ*	Metadata only (no abstract)	Particle-bound atmospheric cadmium (concentration and deposition)	China (national)
Liu Beijing autumn-winter $PM_{2.5}$ oxidative potential source predictors	Autumn-winter variability and source-related predictors of $PM_{2.5}$ oxidative potential in urban Beijing	*Atmos Environ*	Metadata only (no abstract)	$PM_{2.5}$ oxidative potential (assay not stated) and source tracers	China (Beijing)
Liu $PM_{2.5}$ sources mixing state Chengdu Kangding altitude	Altitudinal and seasonal differences in PM(2.5) sources, secondary formation, and particle mixing states in the basin-plateau transition zone of Southwest China.	*Environ Monit Assess*	Source apportionment	Filter $PM_{2.5}$ chemistry (ions, carbon) and single-particle mass spectrometry	China (Chengdu and Kangding, Sichuan)
Lu street-view greenery street canyon PM concentrations	Association between urban greenery and particulate matter concentrations in street canyons: a local-scale study based on street view imagery	*Environ Sci: Atmos*	Field measurement	Street-canyon PM concentrations (metric not stated in abstract)	Not stated in abstract
Magalhaes extreme rainfall landslides PM_{10} $PM_{2.5}$ Sao Sebastiao	Can extreme rainfall associated with landslides affect air quality? A case study in a coastal city	*Environ Monit Assess*	Monitoring-record analysis	Hourly PM_{10} and $PM_{2.5}$ (regulatory station), Dec 2021-Feb 2025	Brazil (Sao Sebastiao, Sao Paulo State)
Mai population-weighted $PM_{2.5}$ O3 China 2015-2024 preprint	Population Weighted Ambient $PM_{2.5}$ and O3 Concentrations in China from 2015 to 2024	*Research Square (preprint)*	Exposure model (secondary data)	Annual 1 km CHAP $PM_{2.5}$ and O3 weighted by WorldPop	China (national)
Makacha PRECISE personal ambient $PM_{2.5}$ monitoring framework pregnancy SSA	An integrated climate-structured personal and ambient air pollution monitoring framework embedded within a pregnancy and postpartum cohort in Sub-Saharan Africa	*Front Glob Womens Health*	Instrument evaluation	Minute-resolved personal and fixed-site $PM_{2.5}$ (wearable LCS)	Sub-Saharan Africa (Gambia, Kenya, Mozambique)
Mulena dust intrusion regimes north-central Argentina vertical structure optical properties	Dust intrusion regimes over north-central Argentina: Vertical structure, optical properties, and transport pathways	*Atmos Res*	Metadata only (no abstract)	Mineral dust aerosol vertical profiles and optical properties	Argentina (north-central)
Nolles IGP vs NCP $PM_{2.5}$ pollution forming efficiency CAMS	Indo-Gangetic Plain experiences higher aerosol pollution than North China Plain despite lower emissions	*Environ Sci Pollut Res Int*	Reanalysis / inventory analysis	CAMS EAC4 $PM_{2.5}$ vs CAMS-GLOB-ANT emissions	South Asia and China (IGP vs NCP)
Park Paris UFP number emissions mass-number conserving	Reconciling mass and number emissions through particle size: Implications for urban ultrafine particle modelling	*J Hazard Mater*	Chemical transport modelling	Size-resolved particle number (UFP) from WRF-CHIMERE/MUNICH/SSH-aerosol	France (Greater Paris)
Perez-Vizcaino PM_{10} sources Riotinto mining multi-time-resolution PMF	Characterization of PM(10) sources in the Riotinto copper mining district (SW Spain) using multi-time-resolution receptor modelling.	*J Hazard Mater*	Intensive field campaign, PMF source apportionment	Hourly XRF elements and Aethalometer BC + daily filter PM_{10} chemistry	Spain (Riotinto mining district)

First author	Focus	Journal	Design	Metric	Region
Savolainen road traffic emission factors impact factors Helsinki street canyon	From direct emission factors to inverse and indirect impact factors of road traffic: influence of emerging and unregulated pollutants	*Atmos Chem Phys*	Field measurement	Particle number and size distribution, BC, PM_{10}, $PM_{2.5}$, PM chemical components, NOx, CO, VOCs, small ions	Finland (Helsinki street canyon)
Tripathi satellite-reanalysis sulfate aerosol India ground measurements	Improving Satellite-Reanalysis Sulfate Aerosol Estimates over India Using Ground Measurements	*Environ Pollut*	Metadata only (no abstract)	Sulfate aerosol (satellite/reanalysis vs ground)	India
Tsai mineral dust biomass burning aerosol convergence East Asia western North Pacific	Convergence of mineral dust and biomass-burning aerosols over East Asia and the western North Pacific	*Atmos Environ*	Metadata only (no abstract)	Mixed mineral dust and biomass-burning aerosol (transport)	East Asia
Xiao NCP Sichuan haze reduced-nitrogen NOC oligomerisation $PM_{2.5}$	Aqueous Carbonyl-Amino Aggregation Drives Atmospheric Nitrogen-Containing Oligomer Formation in China Haze	*Environ Sci Technol*	Field measurement	$PM_{2.5}$ nitrogen-containing organic compounds (ESI+ HRMS)	China (North China Plain, Sichuan Basin)
Zhao $PM_{2.5}$-O3 co-pollution episodes BTH YRD	Episode analysis of the characteristics and causes of $PM_{2.5}$–O3 co-pollution in Beijing-Tianjin-Hebei and Yangtze River Delta, China	*Atmos Pollut Res*	Metadata only (no abstract)	$PM_{2.5}$ and O3 co-pollution episodes (monitoring data)	China (Beijing-Tianjin-Hebei and Yangtze River Delta)
de la Cruz automated forced air injection cordwood stove $PM_{2.5}$ particle size	Less Mass of $PM_{2.5}$ Emissions, Smaller Particles: The Trade-Off of Automated Forced Air Injection in a Residential Cordwood Stove	*Atmos Environ*	Metadata only (no abstract)	$PM_{2.5}$ mass emission and particle size distribution	Not stated
Burden, policy & mitigation					
Beveridge HeatRisk AQI cardiovascular action algorithm	HeatRisk and air quality risk recommendations: A narrative review and action algorithm for the care of people at risk for and living with cardiovascular disease	*Am J Prev Cardiol*	Narrative review	Air Quality Index ($PM_{2.5}$-driven) alongside NWS HeatRisk	United States
Brunori Rome urban vineyard grapevine leaf PM interception	Plant-Atmosphere Interactions in Urban Vineyards of Rome: A Digital Phenotyping Approach for the Functional Selection of Grapevine Cultivars	*Plants (Basel)*	Field measurement	Leaf-surface PM particle counts (SEM, particles/mm2)	Italy (Rome)
Cantu air quality across the lifespan nursing response review	Breathing Unequal Air: Global and US Air Quality Across the Lifespan and the Nursing Response.	*Nurs Clin North Am*	Narrative review	$PM_{2.5}$, O3, NO2 (reviewed)	Global / multi-region (with United States focus)
He ozone-dominant air pollution risk ageing inequality China counties	A nationwide effect-based assessment of ozone-dominant air pollution risk patterns and ageing-related inequality in China	*Ecotoxicol Environ Saf*	Ecological / registry analysis	Cold-season $PM_{2.5}$ and warm-season O3 (multi-source county exposures)	China (813 counties)
Idowu medical waste incinerator airborne heavy metals Monte Carlo Nigeria	Monte Carlo-Based Probabilistic Health Risk Assessment of Airborne Heavy Metals from a Medical Waste Incinerator in Southwestern Nigeria	*Research Square (preprint)*	Health impact model	PM-bound heavy metals (passive glass-slide deposition samplers)	Nigeria (Ido-Ekiti)
Jiang PLA melt-blown nonwoven filtration durability humidity airflow	Service Durability of PLA Melt-Blown Nonwovens: Roles of Humidity and Airflow Velocity in Degradation and Filtration Performance	*Langmuir*	Chamber / laboratory	Filtration of particulate matter by PLA melt-blown media	Laboratory
Liang emergency haze controls causal ML synthetic control Shijiazhuang	Evaluating Emergency Emission Controls During an Unprecedented Regional Haze Event Using Causal Machine Learning	*Environ Sci Technol*	Augmented synthetic control + weather normalisation	Monitor $PM_{2.5}$, PM_{10}, SO2, NO2, CO; $PM_{2.5}$ source apportionment	China (Shijiazhuang)
Polwiang Bangkok Chiang Mai burning control DLNM GAM $PM_{2.5}$ regimes	Differential sensitivity of urban and biomass-dominated $PM_{2.5}$ regimes to burning control: A distributed-lag generalized additive model analysis of Bangkok and Chiang Mai	*Atmos Pollut Res*	Metadata only (no abstract)	Daily $PM_{2.5}$ (urban vs biomass-dominated regimes)	Thailand (Bangkok, Chiang Mai)
Prieto Perez Huszar future central European air quality emission changes	The role that changes in emissions play in future central European air-quality	*Atmos Environ*	Metadata only (no abstract)	Future PM and O3 under emission scenarios (presumably)	Central Europe
Scott equine air quality index Delphi management	Experts recommend management strategies for horses during poor air quality	*J Am Vet Med Assoc*	Delphi consensus	AQI categories ($PM_{2.5}$-driven, wildfire smoke)	USA (expert panel)
Warminski SRC pellets boiler ESP PM CO emissions	Short-Rotation Coppice Pellets as Biofuels in a Modern Boiler with an Electrostatic Precipitator: Particulate Matter and Carbon Monoxide Emissions	*Molecules*	Laboratory measurement	Flue-gas total PM mass (mg/m3) with and without ESP	Not stated in abstract

First author	Focus	Journal	Design	Metric	Region
Zhang Jiaxing inventory co-control screening $PM_{2.5}$ GHG	Inventory-Based Co-Control Screening of Air Pollutants and Greenhouse Gases in a Typical Yangtze River Delta City	*Atmosphere*	Emission inventory	Inventory $PM_{2.5}$ (with NOx, SO2, VOCs, NH3) vs CO2-eq	China (Jiaxing, Yangtze River Delta)
Zhang early-onset female lung cancer burden GBD 2021	Global Burden and Trends of Early-Onset Tracheal, Bronchus, and Lung Cancer Among Women From 1990 to 2021.	*Asia Pac J Clin Oncol*	Burden estimation (GBD)	Ambient particulate matter pollution (GBD risk factor)	Global / multi-region (GBD 2021)
Zhang respirable magnetite emissions coal-fired power China	Emissions of Respirable Magnetite from the Coal-Fired Power Sector in China: Spatial Inequalities and Mitigation Pathways	*Environ Sci Technol*	Emission inventory	Respirable magnetite (Fe3O4) particle emissions	China (national, provincial)
Other clinical endpoints					
Bottini antibiotic resistance Ecoli Salmonella $PM_{2.5}$ ecological	Environmental, socioeconomic, and livestock-related factors associated with antibiotic resistance in Escherichia coli, Shigella spp., and Salmonella enterica	*One Health*	Ecological	Country-level $PM_{2.5}$ exposure	Global / multi-region (102 countries)
Jiang air pollution meteorology hand foot mouth disease Shenzhen DLNM	Ambient air pollution and meteorological factors associated with hand, foot, and mouth disease risk in Shenzhen, China: a time-series study	*Front Public Health*	Time-series	Daily city-wide $PM_{2.5}$ and PM_{10} (fixed monitors)	China (Shenzhen)
Lin $PM_{2.5}$ frailty trajectories CHARLS	The association of particulate matter 2.5 exposure with frailty trajectories in middle-aged and older Chinese adults and the potential modification by lifestyle and socioeconomic status	*J Occup Environ Med*	Prospective cohort	Satellite-derived $PM_{2.5}$ (Dalhousie/ACAG surface, per 10 $\mu g/m^3$)	China (national, CHARLS)
Lin air pollution cervical cancer recurrence mortality Fudan cohort	Long-Term Exposure to Ambient Air Pollution and Cervical Cancer Outcomes in Chinese Women: A Prospective Study on Recurrence and Mortality Risks	*Environ Health (ACS)*	Prospective cohort	Residential $PM_{2.5}$ and PM_{10} from high-resolution spatiotemporal models (post-surgery)	China (Shanghai, single hospital)
Wen landscape fire $PM_{2.5}$ renal hospitalisations six-country time-series	Landscape Fire-Sourced $PM_{2.5}$ and Cause-Specific Renal Hospitalizations: A Multicountry Two-Stage Time-Series Analysis of 12.5 Million Admissions	*Environ Sci Technol*	Multi-location time-series	Landscape-fire-sourced vs non-fire $PM_{2.5}$ (per 10 $\mu g/m^3$, lag 0-5)	Multi-country (Australia, Brazil, Canada, Chile, New Zealand, Thailand)

Provenance and method

Period **18 – 24 October 2026**, Sunday to Saturday, **seven daily issues**, **119 records**, **119 unique DOIs**, no record without a DOI. This rollup is **aggregation only**: it re-reads `state/corpus/*.json`, `state/metrics.csv` and `state/trials.json` over the period and **queries no literature API**; every record was gated by `check_dois.py` at daily-issue time. Tier mix **21 A, 43 B, 55 C**. The 23 and 24 October dailies and this rollup were built on the **30 October recovery run** (the scheduled runs for 23–29 October did not complete); no record is held. Effect estimates are transcribed verbatim from the daily issues on heterogeneous contrasts and are **not** meta-analysed. Abstracts remain the copyright of their respective publishers.